

Brokers Who Do Not Bridge: When Assessment, Not Access, Shapes Broker Advantage

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Abstract

Structural theories describe economic brokerage as an opportune but fragile position bridging gaps between otherwise disconnected actors. In that view, brokering can undermine trust or eliminate the gaps that generate its value, and brokers need supporting institutions or deeper interpersonal ties to stabilize their role. Yet many corporate actors in positions of brokerage, from Visa to Amazon, have risen in market power, profit, and prominence even as networks transformed around them. When can brokering strengthen rather than undermine a broker's advantage?

I reconceptualize brokerage as outsourced relational work, in which a broker who assumes no inventory risk constructs viable matches between parties who cannot easily find or evaluate one another. Repeated matchmaking generates an informational byproduct that the broker can accumulate: knowledge of what makes pairings successful across the market, unavailable to clients who observe only their own dealings. An agent-based model of a matching market identifies conditions under which this informational advantage emerges and persists. Brokerage can provide assessment rather than access: clients outsource matching even when they can reach counterparties themselves, because the broker's predictions are valuable. Counterintuitively, the broker can become highly central in regimes where it engages in little bridging behavior: broker centrality is a product, not a primary source, of matchmaking quality.

While the model explores the dynamics of the broker's informational advantage, I theorize how this advantage can then enable the broker to become a principal selling the resource it once intermediated or monetizing data and analytics. This is *transient brokerage*, a process that highlights a broker's growing power rather than its fragility.

I Introduction

Intermediation in economic relations—the intervention of a third party to mediate or facilitate exchange between others—can take many forms, including that of brokerage. Brokers are intermediaries that do not assume inventory risk: they act as matchmakers, connecting others.

Brokerage is pervasive in a wide variety of markets and remarkably persistent across time and space. Brokers find themselves in an advantageous but delicate position. Unlike dealers and wholesalers, brokers do not sit on an inventory that they need to manage and can liquidate in case of difficulty. Unlike market makers, they keep offers and price information private (Rust and Hall 2003). They operate at the intersection of social groups (Burt 1992, 2004) and rely on secret information. As a result, they often lose the trust and legitimacy needed to continue their activities. Prominent theories of brokerage have described it as an opportune but inherently fragile role that often gets “stabilized” to avoid collapsing, through the social isolation of brokers, their organizational grafting, or their capture by one side (Stovel, Golub, and Milgrom 2011).

If that is the case, then how have so many corporate actors in positions of brokerage, from Visa to Amazon, managed to rise in market power, profit, and prominence even as surrounding networks changed in ways that could have undercut their brokerage positions?

Recent work by Ronald S. Burt, building on the tradition of structural-hole brokerage theory, has shown one important reason why the broker fragility narrative fails: many bridging ties are not weak and fragile, but rather strong high-trust ties forged through interpersonal history, and they provide a robust and enduring advantage to the broker who relies on them (Burt 2026; Burt and Oppen 2026). However, this is not the full story.

Some brokers leverage their role to become principals rather than remain only intermediaries. In so doing, they stabilize their advantage and consolidate power while brokering becomes secondary or even disappears entirely. I call this *transient brokerage*, a phenomenon that current theories of brokerage do not satisfactorily predict or explain. Transient brokerage highlights how brokering sometimes declines or disappears because of the broker’s power, not its fragility.

To explain transient brokerage, this article redefines brokerage as *outsourced relational work*: the broker constructs viable matches between parties who cannot easily find or evaluate each other. This relational work generates an informational byproduct that the broker can leverage. Structural position provides the access that feeds learning, but while each successful match between previously unconnected parties can erode the broker’s structural advantage by densifying the network (Burt 2000, p. 356), it also strengthens the broker’s informational position. The broker accumulates knowledge of what makes pairings successful.

In that way, the broker converts structural capital into informational capital through the act of brokering. This is particularly true for corporate brokers, who operate at scale and whose relational work benefits from legal privileges conferred by the corporate form. Depending on the market in which the broker operates, the nature of the matching problem, and the relations between the actors involved, the broker’s information advantage can support a transition from intermediation to capture. The broker dismantles the relational package that was negotiated with its principals and converts vicarious relational work into personal economic advantage. If successful, **the broker becomes a principal selling the resource it once intermediated or monetizing data and analytics**. *Corporate* relational work is a particularly advantageous basis for such capture. This explains the rise in prominence of some firms acting as brokers.

This article develops a theory of brokerage as outsourced relational work and explains how the resulting informational advantage can enable a transition from intermediary to principal. An agent-based model formalizes one part of this argument, illustrating how repeated matchmaking can generate informational and structural advantages and producing propositions for future empirical research. The article addresses three questions about brokerage:

- (i) What value does a broker provide?
- (ii) How does brokering alter its informational and structural position?
- (iii) Under what conditions can the broker convert its advantage into resource capture or data enclosure?

I develop an agent-based model (ABM) of brokerage in a general matching market to formalize and illustrate how brokers can create and preserve an informational advantage through repeated matchmaking. The simulations show how several simple elements interact to produce non-obvious dynamics, complicating the expectation that brokers densify networks around them and erode their own position. The ABM results suggest that brokerage can provide assessment rather than access, and that clients may continue outsourcing even when they can reach counterparties themselves. Successful matchmaking can increase the broker’s structural prominence. Centrality, or bridging position, is distinct from bridging behavior: a broker can be highly central while doing little or no bridging. The ABM results illustrate this and recast betweenness centrality not only as a source of brokerage opportunity, but also as a product of the kind and quality of a broker’s relational work.

The rest of this article proceeds as follows. Section II reviews structural and processual theories of brokerage and their blind spots. Section III reconceptualizes brokerage as outsourced relational work and traces how this work is transformed when performed at scale by corporate actors. Section IV presents the ABM and its results illustrating how a broker produces and sustains its advantage through this work. Section V builds on this foundation to theorize one possible trajectory that can follow from the broker advantage, transient brokerage, and the conditions under which a broker can become a principal. Section VI discusses scope conditions and implications for the political economy of corporate intermediation, and Section VII concludes.

II Current Theories of Brokerage Do Not Explain Capture by the Broker

Brokerage tends to emerge in markets where information is scarce or complex, participants are few, transactions are intricate, or institutions are poorly developed (Stovel, Golub, and Milgrom 2011). These conditions make participants hard to find, resources hard to value, or transactions hard to structure. They also make the intermediation work of a broker valuable. Real estate, mortgage, stock, and commodity markets are common examples of markets with many active brokers. Brokerage may also be helpful in circumstances where tight-knit, closed groups monopolize resources or information, and outsiders cannot easily access them (Stovel and Shaw 2012, p. 140). Brokerage can maintain or reinforce group boundaries, or it can blur or collapse them (Stovel and Shaw 2012), with consequences for social segregation, social integration, and relations between groups or communities (Galster and Godfrey 2005).

Past research has proposed two conflicting sets of definitions of brokerage (see Erikson 2013, for a review): narrower formalist or structuralist definitions (Burt 1992; Gould and Fernandez 1989; Granovetter 1973; Marsden 1982), and broader processual definitions (Obstfeld, Borgatti, and Davis 2014; Rossman 2014; Spiro, Acton, and Butts 2013; Stovel and Shaw 2012).

A Formalist and Structuralist Definitions

Early brokerage research examined the structure of networks and the positions of actors to identify brokerage forms, categorize them, and determine their consequences (Burt 1992, 2004; Gould and Fernandez 1989; Granovetter 1973; Marsden 1982; Simmel 1964, 1971). Brokerage opportunities arise when there are gaps in a social network, such that some nodes are close (in degrees of separation) yet disconnected (Burt 1992, 2004; Marsden 1982). Brokerage is synonymous with open triads or structural holes: a structure in which a third actor bridges two otherwise unconnected nodes. The third actor mediates flows of resources and information and facilitates transactions between others who lack access to one another (Fernandez and Gould 1994, p. 1457; Marsden 1982, p. 202). Researchers in this tradition generally set aside interactional content to achieve greater generality (Erikson 2013). For instance, Burt (1992, p. 36) was explicit in treating “motivation and opportunity as one and the same.”

Burt (1992) operationalized the structural-hole concept, proposing the measures of network constraint and effective size to quantify how well positioned an actor is for brokerage, which allowed the theory to be tested across a wide range of settings. For instance, in a follow-up study, Burt (2004) investigated the network position of 673 managers in a large electronics company. He found that those who spanned structural holes generated ideas that senior management judged more valuable and were rewarded with better performance evaluations, promotions, and compensation. He suggested that brokers appear more creative because they can take insights that are commonplace in one group and introduce them to other people to whom they appear novel and original (Burt 2004, p. 388).

As such, brokers derive an advantage from their ability to exploit pre-existing information in the network. Because of their position, brokers get privileged access to diverse ideas from across the network. They can see opportunities that others do not, and they can strategically use their position by providing or withholding information (Burt 1992) or using it to sustain their own creative endeavors. Exposed to varied perspectives, they are well-positioned to build a richer understanding, synthesize across domains or outlooks, and develop a unique vision (Burt 2004).

Defining brokerage based on structural holes, however, excludes situations that clearly appear as brokerage to a lay observer. For instance, Alice wants to sell her house and asks a real estate broker to list it. Soon after the real estate broker puts up the ‘for sale’ sign, her neighbor Bob, whom she knows personally, contacts the broker to purchase the house. A broker intermediates the transaction, yet no structural hole existed between buyer and seller.

B Processual Definitions

More recently, some researchers have recast brokering as a set of relational processes (Obstfeld, Borgatti, and Davis 2014; Rossman 2014; Spiro, Acton, and Butts 2013; Stovel and Shaw 2012). Stovel and Shaw suggested that brokerage was “the process of connecting actors in systems of social, economic, or political relations in order to facilitate access to valued resources” (Stovel and Shaw 2012, p. 141; see also Stovel, Golub, and Milgrom 2011). Brokers exhibit two crucial features: they are positioned as bridges across gaps, and they help transfer resources or knowledge across those gaps (Stovel and Shaw 2012, p. 141). This definition remains constrained by social structure but requires brokerage behavior. Hence, there can be structural holes without brokerage—some bridging actors do not transmit anything across holes—but no brokerage without structural holes. In this way, it runs into the same conceptual problem as purely structural definitions.

Obstfeld, Borgatti, and Davis (2014), building on Marsden (1982), proposed a purely processual definition: brokerage is a process by which an actor “alters interaction between two or more parties in a wide variety of triadic structures” (Obstfeld, Borgatti, and Davis 2014, p. 136), meaning that he “influences, manages, or facilitates” the interaction (Obstfeld, Borgatti, and Davis 2014, p. 141). This eliminates any structural requirement beyond the need for three or more actors. It is no longer necessary for the alters to lack access to or trust in each other. There can be structural holes without brokerage, and there can also be brokerage without structural holes: brokerage behavior can, and often does, occur in dense networks.

Previous work on brokerage, particularly in the processual tradition, has also identified categories of brokers depending on the nature of their activities. Middleman, transfer, or conduit brokers facilitate economic transactions without necessarily altering the structure of a local network, like stockbrokers (Stovel and Shaw 2012, p. 146; Spiro, Acton, and Butts 2013; Obstfeld, Borgatti, and Davis 2014, p. 141). Catalyst, moderator, or matchmaking brokers focus on creating lasting ties, like headhunters and party hosts (Stovel and Shaw 2012, p. 146; Obstfeld, Borgatti, and Davis 2014; Spiro, Acton, and Butts 2013). Finally, *coordinators*, such as middle managers or disaster response teams, work to transform existing ties by reconfiguring local networks of interdependence (Spiro, Acton, and Butts 2013, pp. 131–132).

While definitions based on structural holes were very restrictive, Obstfeld and co-authors’ processual definition is sweeping. For instance, they suggested that, under their definition, a manufacturing firm acts as a broker when it buys raw materials from an alter, transforms them into a product, and sells the finished product to another alter (Obstfeld, Borgatti, and Davis 2014, p. 145). Most triadic or polyadic relations, in this account, imply brokerage. Indeed, many descriptions of brokers in the literature are so broad as to invite the charge that a concept that explains everything explains nothing. The most varied social facts have been described as acts of brokerage: mediating conflict (Obstfeld, Borgatti, and Davis 2014), reselling stolen property (Steffensmeier 1986), advocating on

behalf of someone, making sense of the world for someone, and helping connect epistemic communities (Stovel and Shaw 2012, p. 140), to name only a few.

C Shortcomings of Existing Frameworks

Work that relies chiefly on the structural holes idea generally makes clear that transferring knowledge and ideas is a key element of brokering and of the broker advantage. Yet the information under consideration is generally preexisting and ambient, not generated by brokering itself (Burt 2004). Moreover, the structural account leaves the actual work of brokering largely implicit. Structural position indicates places where brokerage is possible, but not whether anyone is doing the work of brokering—work for which structural holes are neither necessary nor sufficient.

Meanwhile, broad processual definitions blur the lines between vicarious and personal relational work. They include under the same umbrella cases where the intermediary deals with others on its own account, and cases where the intermediary works on behalf of others. As such, this perspective does not examine the informational byproduct that results from repeated vicarious work. Information is often simply listed among the resources that can be transferred through a broker, along with material or financial resources (see, e.g. Obstfeld, Borgatti, and Davis 2014, p. 141; Stovel and Shaw 2012, p. 141).

Neither perspective offers a cohesive theory of the role of information in brokerage, even though information shapes every stage of the brokering process. Before brokerage, the distribution of information shapes the opportunity or need for it: brokerage arises partly out of informational inequality, and the broker exploits what it knows that one or both sides do not. During brokerage, the transmission of information is part of the brokerage act itself, as the broker shares enough to make a match viable while withholding enough to preserve its advantage. After brokerage, the distribution of information has been altered, and the work has generated a byproduct the broker retains, which may shape the broker’s opportunities. The circulation of information is at once a determinant of brokerage, a component of the work of brokering, and a consequence of brokerage.

This article reconceptualizes brokering as outsourced relational work and builds on this theoretical move to explain how brokerage sometimes weakens or disappears because a powerful broker captures resources or encloses data to become a principal.

III Outsourced Relational Work and the Broker Advantage

Social actors engage in *relational work* when they try to interpret and match social ties, economic transactions, and media of exchange, to create viable matches that will shape the outcome of their interaction (Zelizer 2012, p. 151, 2005; see also Bandelj 2020, p. 255). These matches and their shared meanings form distinct *relational packages*. For instance, in a marketplace, actors trading directly with one another draw on cultural knowledge to reach a shared understanding of their situation and the ‘rules of the game,’ and to agree on terms, including price (Block 2012, p. 137).

For the broker’s clients, brokerage is *outsourced relational work*. For the broker, brokerage is *vicarious relational work*. In economic markets, this outsourced relational work is usually performed at scale by a *corporation*. This is more than a question of definitions: it shapes the broker’s opportunities, its strategic advantage vis-à-vis its clients, and its trajectory.

A From Personal to *Outsourced and Vicarious Relational Work*

In brokered transactions, brokers take on the work of constructing a shared reality for limited purposes. The core of what brokers do is relational matching (see Zelizer 2012). Brokers are characterized by the relational work they do *on behalf of others*.

For the broker, this vicarious work requires another layer of relational work to negotiate brokerage relations with its clients. Brokerage corresponds with a distinct relational package. Not all intermediation is brokerage. In the relational package of economic brokerage, the broker performs a matching service for others, usually for a fee, commission, or retainer. Brokers bundle various services and amenities as part of that package: they take on the efforts of search and negotiation, they provide access to otherwise inaccessible counterparties, they share with their clients their expertise and the benefits of the private information they hold about quality, pricing, and availability, and they document and facilitate transactions, which often require logistical, bureaucratic, or regulatory work.

The middleman, catalyst, and coordination brokerage described in Section II are not exclusive categories of brokers, but rather, components of the work of brokering, which involves a bit of each in varying proportions. An important part of brokers' relational matching is to facilitate interaction (middleman brokerage), but this work also directly impacts interpersonal relations and local networks, often in an enduring way, by creating new and lasting ties (catalyst brokerage) or by reorganizing local networks (coordination brokerage).

Brokerage inherits the dual character of relational work, which can both foster and reduce informational asymmetry when constructing shared understandings while maintaining strategic opacities (Block 2012, p. 138). In performing relational work on behalf of others, the broker simultaneously clarifies and obscures: it transmits enough information to make a transaction legible and a match viable and conceals enough to justify its fee and preserve its advantage. The same opacity can do social as well as economic work. Rossman (2014) shows how brokerage functions as obfuscatory relational work that manages the taboos around disreputable exchange. Outsourcing relational work provides cover to parties in morally fraught matches. What the broker withholds is also the raw material of its later advantage: information that it withholds, accumulates, and, under the right conditions, eventually monetizes.

Vicarious work relates to entities or resources that are *not* under the broker's personal ownership and control, and the shared understanding of the parties is that the broker is *not a principal* in the brokered transaction (see, e.g. Stovel, Golub, and Milgrom 2011, p. 21326). Brokers generally have limited personal involvement in brokered interactions beyond their role as facilitators. The broker is a supporting actor, not a main character. Intermediaries who have full control over traded resources occupy a very different position in terms of risk, power, and vulnerability.

Another way to put it is that in economic markets, brokers are intermediaries that *do not assume inventory risk*. That is, if Bob helps Alice and Charlie conclude a sale, Bob is a broker. If Bob buys from Alice and eventually leases to Charlie, he is not. For instance, a real estate broker using her market knowledge to buy houses in up-and-coming neighborhoods to rent to families no longer acts as a broker. Instead, the relational work she performs is personal: matching herself to sellers and renters.

B From Individual to *Corporate* Relational Work

The definition above applies naturally to describe individual brokers working with individual clients, such as a real estate agent, an insurance broker representative, or a matchmaker. Much of the work of brokerage, however, occurs at the corporate, organizational, or institutional level, where brokers are firms, and clients may also be.

To understand what happens when relational work is outsourced to corporations, we must first discuss how corporations perform relational work in the first place, for themselves as organizational social actors. Relational work scholarship initially focused on the intimate sphere and interpersonal relationships (see Bandelj 2020, for a review). But relational work also takes place in the economic sphere and at the organizational level. The relational work tradition has acknowledged the organizational question without fully resolving it. Bandelj (2012) noted that relational work between

representatives of organizations solidifies into macrolevel structures, while Zelizer (2012, pp. 156–157) catalogued instances of relational work by firms in the field of labor relations. Bandelj’s (2020) review of “scaling up” relational work surveyed studies operating at the organizational level, but they mainly focused on what people do with and to each other within organizations, not on the corporate form itself as a vehicle for relational work.

Firms must conduct relational work to manage contingent, fleeting ties and longer-term relationships with workers, consumers, suppliers, lenders, investors, bureaucrats, and so on. The properties of the corporate form matter to this work and make it distinct. The firm is not simply a site where interpersonal relational work occurs; nor is it a singular, scaled-up actor that performs relational work as an individual does.

Firms are groups of people in designated roles, following norms and practices, supported by procedures, interfaces, and infrastructures, and doing the everyday work of managing the firm’s relationship with its environment “at the coalface” (Gray and Silbey 2014; Huising and Silbey 2018). The mining metaphor directs attention to the frontline of the organization, where people enact its dealings with the outside world and translate organizational imperatives into routine practice. Relational work, like everything the corporation does in the world, is enacted through the people, procedures, infrastructure, and algorithms working at the coalface. Because of its collective and distributed nature, it is never visible in its entirety to any of the people involved.

Most firms are also legally organized as corporations. The corporation is a political entity with emergent capacities for coordination, information processing, and strategic action that exceed the sum of its parts (Ciepley 2013; Farrell and Newman 2023). There is a fundamental asymmetry between corporate actors and natural persons (Coleman 1982, 1990). The corporation aggregates information across thousands or millions of interactions, while each natural person sees only their own. The corporation can persist indefinitely, while individual relationships are transient. The corporation coordinates strategically across its full portfolio of relationships, while each counterparty acts in isolation. As a result, when a corporate actor interacts with natural persons, the corporate actor has structural and informational advantages.

These asymmetries are not accidental byproducts of size: they are legally constituted features of the corporate form. The corporation is one of the core modules of the legal code of capital (Pistor 2019). Corporate law endows the firm with durability that outlasts any individual relationship, priority that ranks its claims above those of other parties, and universality that makes those claims enforceable against the world, so that it can become a legal technology for concentrating and protecting accumulated advantages and leveraging them for productive purposes. These are the conditions under which corporate relational work takes place: it is inherently asymmetric, because of the different scales and material and informational resources of corporate and natural persons.

C Outsourcing Relational Work to Corporations: The Corporate Brokers

When a client outsources relational work to a corporate broker, the corporate broker performs that work through people and practices at the coalface, each working in some small way to define relational packages, perform appropriate matches, and shape information flows. The corporate form reorganizes vicarious relational work, distributes it across roles, and embeds it in routines and procedures. To orchestrate this, corporate brokers develop an organizational apparatus comprising internal and client-facing roles, information architectures and databases, contractual templates, and interfaces.

Saying a brokerage firm does relational work on behalf of its clients means that in sales meetings, in contract negotiations, in the design of user interfaces, in the defaults embedded in terms of service, and so on, people relying on organizational procedures are working to define the features of its clients’ relationships, what transactions are appropriate to them, and what media may govern the exchanges. Just like for its own relational work, the corporation has a capacity for coordination, information

aggregation, and temporal persistence that makes the vicarious relational work of corporate brokerage qualitatively different from what any individual could accomplish alone.

The corporate broker’s relation with its clients is structurally asymmetric. With human clients, corporate and natural persons bring fundamentally different legal, informational, and temporal architectures to the relationship. Even with corporate clients, however, the asymmetry remains: the broker coordinates vicarious relational work strategically and benefits from cross-market information, while the client encounters it piecemeal. The matching of relations, transactions, and media moves from a pure negotiation between principals with comparable standing, who learn from each transaction, toward an administered process in which the corporate broker sets the terms, explicitly or behind the scenes, and captures the information that each pairing generates.

The data infrastructure of the corporate broker includes data taxonomies, client categories, risk models, algorithmic sorting procedures, and so on. This infrastructure does not merely facilitate transactions but actively constitutes relational packages by determining what transactions are possible, and what categories they should fall into. Classification systems and information infrastructures, like those of the corporate broker, are not neutral descriptions of the world; they actively organize social relations (Bowker and Star 1999; Fourcade and Healy 2024). The corporate broker’s information infrastructure, much like a corporate rating agency’s credit-scoring system or a platform’s matching algorithm, sorts counterparties into categories the broker defines and uses those categories to allocate opportunities according to the broker’s objectives. The impact of the broker’s information infrastructure is magnified because the corporate broker works at scale, while individuals cannot usually design categories that simultaneously structure millions of relationships.

D The Broker Advantage

Why do people and firms outsource their relational work to brokers? To have the opportunity to broker, a participant in a matching market must hold either a *structural* or an *informational* advantage. A structural advantage arises from network position and can be used to provide *access*: the broker bridges by bringing together parties that were not previously connected. An informational advantage supports *assessment*: the broker evaluates prospective matches better than its clients can. The broker’s service is therefore valuable when it helps clients reach counterparties they could not otherwise find or better evaluate counterparties already available to them.

Assessment involves two components: evaluating the general quality of the goods or counterparties involved and evaluating the quality of the specific pairing or relational package being considered. These components also correspond to two possible channels for the broker’s information advantage: *attributional*, about how parties’ attributes translate into individual quality, and *relational*, about which counterparties complement each other. Both are self-reinforcing, as the broker accumulates the knowledge required for assessment through the relational work it performs, and that learning makes it better at the work. The broker observes outcomes from many pairings across many principals, while each principal observes only its own. This gives the broker both a cross-market view and higher data volume.

The broker’s information work can be informal and internal, as with a real estate broker who knows the local market, has an intuition for the right asking price, and holds social ties to likely buyers. It can also be formal and bureaucratized, as with a corporate mortgage broker that follows tables and formulas to determine which interest rate may produce successful matches between borrowers and lenders. The latter records and retains what the former holds only in personal experience, and this difference reflects that between individual and corporate brokerage.

Important models in the economics literature have focused on the attributional channel and characterized the broker’s role primarily as quality certification (Li 1998) or expert screening (Bethune, Sultanum, and Trachter 2022): the broker identifies which goods or counterparties are high quality.

In that view, the broker is an appraiser whose cross-market experience helps it assess the general quality of counterparties more accurately than individual principals can. The relational-work view of brokerage suggests that the broker’s value lies also, and often primarily, in understanding complementarities between counterparties and shaping relational packages accordingly. The broker is a relational worker and a matchmaker whose advantage comes from the ability to make pairings and to know which pairings will likely succeed, an ability that improves with repeated matchmaking.

The formal illustration that follows isolates one upstream part of the theory: a repeat-player broker observes outcomes across clients and uses the resulting information to organize subsequent matches. It examines how assessment, access, outsourcing, and network position can evolve together. It does not model the corporate form or the subsequent transition to resource capture or data enclosure.

IV How Brokers Accumulate Informational and Structural Advantage: A Formal Illustration

A Model and Methods

I use an agent-based model (ABM) to formalize the theory. The model is a formal illustration, not a test of the theory; it serves as a bridge from theory to testable propositions. Complete pseudocode and specifications appear in Appendices A and B. The public git repository (<https://github.com/m-laprise/brokerage-abm>) contains the Julia code and scripts needed to reproduce the simulations, statistics, figures, and manuscript text.

The model isolates the informational consequences of outsourced relational work. Principals learn only from their own matches, whereas the broker pools information from matches it mediates across principals. Learning general quality represents attributional assessment, while learning complementarity represents relational assessment. The broker represents the information-aggregating function of corporate brokerage, not its internal organization, and the model does not simulate the later transition to resource capture or data enclosure.

Market and match output. Principals are the market participants seeking partners. Many principals and one broker occupy an undirected network. The model represents match quality as match output. Each principal has a fixed vector of observable characteristics, which defines its type. For any potential pair, expected match output is determined by the two principals’ types and by fixed features of the matching environment drawn at the start of each simulation run. It combines general quality, which depends on each principal separately, and complementarity, which depends on how their types interact (Appendix B §2).

Before they are combined, the two components are centered and scaled to have the same variance. The parameter ρ sets their relative contributions to variation in expected match output: general quality contributes a fraction ρ , and complementarity contributes $1 - \rho$. Thus, $\rho = 0$ represents pure complementarity and $\rho = 1$ represents pure general quality. The parameter δ controls how sharply complementarity changes across a boundary in type-pair space: it multiplies complementarity by $1 + \delta$ on one side and $1 - \delta$ on the other. Larger values make otherwise similar pairs on opposite sides of the boundary differ more in expected output, making complementarity harder to learn. A common normalization holds the average within-market variance of expected match output constant as ρ and δ vary. Because δ affects only complementarity, it has no effect when $\rho = 1$.

When a pair matches, its realized output equals its expected output plus an independent, mean-zero random shock. This shock represents match-specific variation that neither the principals nor the broker can anticipate. Supplementary Figures S1–S3 visualize the type distribution, expected match output, and its effective dimensionality.

Search, matching, and learning. Each period, principals seek pairwise matches either through self-search or through the broker (Appendix A §§5–7; Appendix B §§6–7). They choose one channel using satisfaction with past realized output net of search costs or broker fees. Self-search considers existing neighbors and a random sample of other principals. The broker considers principals in its standing roster, its persistent list of contacts, and those currently using its services. Principals estimate expected output from their own match histories. The broker pools the matches it has mediated and uses both parties’ characteristics, without privileging either party. Each uses a small neural network for these predictions.

Estimated output determines how candidates are ranked. For previously matched neighbors, self-search instead uses the pair’s average past realized output. Tied estimates are ordered uniformly at random. Principals make offers to the highest-ranked candidates whose estimated output exceeds the reservation threshold, the minimum required for an offer. All offers enter a common market. A completed match produces realized output and creates a direct tie that persists until either principal exits. Each principal observes the partner’s type and realized output from its own matches; the broker observes both types and realized output from every match it mediates. At the end of each period, every principal exits independently with probability η and is replaced by a new entrant.

Measurement. The outsourcing rate is the share of requested match positions routed through the broker. A brokered placement is classified as access when it creates a direct tie between previously unconnected principals and as assessment when it reuses an existing tie. The access fraction is the share of brokered placements classified as access. It measures network-expanding brokerage, not whether a counterparty was impossible to discover through self-search. The output gap is the difference in mean realized output between brokered matches and self-search matches. Broker betweenness centrality measures the share of shortest paths between principals that pass through the broker (Brandes 2001; Freeman 1977). Supplementary Figures S4–S6 reproduce the structural analyses using Burt’s constraint and effective size.

Prediction quality is evaluated each period using randomly sampled potential partners. These evaluations do not update either model, although a sampled pair may have matched previously. Predictions are compared with expected output, without a random shock. Rank correlation is Spearman’s correlation between the candidate ordering produced by predicted output and the ordering produced by expected output; tied predictions are ordered uniformly at random before the correlation is calculated. The ranking advantage is the broker’s rank correlation minus the principals’ rank correlation. R^2 measures accuracy in predicting expected output levels, but is diagnostic only. It can become extremely negative when expected output varies little across the sampled candidates and is not used in any choice made by the principals or broker (Appendix B §8).

Simulation design and analysis. At the baseline, $N = 1,000$ principals interact for $T = 500$ periods with $\rho = 0.5$, $\delta = 0.5$, and $\eta = 0.02$. One-at-a-time (OAT) sweeps vary the matching problem, turnover, market size, reservation threshold, initial network, broker roster, and self-search pool. Four two-parameter grids cross ρ with η , ρ with the reservation threshold, η with the reservation threshold, and ρ with δ (Appendix B §9). I refer to each unique model configuration as a regime. Together, the sweeps and grids contain 80 regimes. There are 20 random seeds per regime and 50 at the baseline (1630 runs total).

Neural-network performance depends on the learning rate and the number of training updates, so I calibrated these settings before running the main simulations. I compared candidate settings for principals and the broker separately, considering ranking performance on randomly sampled candidates, stability over time and across random seeds, and computational time. I then confirmed the selected settings together using five additional calibration seeds that were not used in the reported simulations. The candidate settings and seed-level results are preserved in the replication code.

To assess whether the findings depend on neural-network learning, I repeat the complete design with Ridge regression. In the unrestricted Ridge specification, principals use candidate characteristics, while the broker uses both parties’ characteristics and interactions between them. Ridge penalties were calibrated at the baseline using ranking performance on randomly sampled candidates and random seeds excluded from the reported simulations. Three additional Ridge experiments, conducted at the baseline and across the $\rho \times \delta$ grid, restrict the broker by giving it the same number of observations as a typical principal, retaining only one party’s characteristics from each match, or omitting interactions between the two parties’ characteristics. The complete calibration and experimental procedures are preserved in the replication code.

Ensemble means are averages across random seeds. Late-window means first average periods $t \in [481, 500]$ within each random seed and then average those seed-level values; early-window means use $t \in [50, 70]$. Cross-regime summaries give each regime equal weight. Intervals for differences are calculated within each random seed before being combined across seeds. Time-series figures begin at $t = 30$ to omit the initialization transient, without changing the simulations or reported averages. Monte Carlo convergence was assessed in two ways. For each regime and reported outcome, I calculated a two-sided 95% confidence interval for the mean across independent random seeds using the t distribution. I also compared nested estimates based on the first 10 versus all 20 seeds outside the baseline and the first 20 versus all 50 seeds at the baseline. Across the 14 non- R^2 figure outcomes, the median across regimes of the confidence-interval half-width divided by the absolute estimate based on all available seeds ranged from 0.3% to 8.3%. The median across regimes of the absolute difference between the nested estimates, divided by the same full estimate, ranged from 0.1% to 4.8%. These relative quantities are not reported when the full-sample interval includes zero because division by an estimate near zero is unstable; those cases are assessed in the outcome’s original units.

B The Broker Primarily Provides Assessment, Not Access

Across the tested regimes, the broker is usually the dominant channel in the market. The outsourcing rate as a share of demand is 0.97 at the baseline (late mean, up from an early mean of 0.78) and has a median of 0.96 across regimes. It exceeds one-half in 78 of 80 regimes; the remaining regimes all have no turnover. Yet the broker primarily assesses options within the existing network rather than providing access beyond it. The access fraction (the share of brokered placements between principals with no pre-existing tie) has a late mean of 0.09 across regimes, down from 0.17 early, and never exceeds 0.18. It declines over time in all regimes, and its paired 95% interval is below zero in every regime. Most brokered matches therefore connect principals already tied to each other. Figure 1 and Table 1 summarize these baseline dynamics and the main measures.

At the baseline, the broker’s realized-output advantage develops as the market evolves: the output gap (broker minus self-search) changes from -0.39 early to $+1.71$ late. Across regimes, mean realized output is higher through the broker channel in 66 of 80 regimes, and the late output gap averages $+1.14$. Paired 95% Monte Carlo intervals are above zero in 65 regimes and below zero in 14. The negative gaps occur in 14 regimes, all with no or nearly no turnover ($\eta \in \{0, 0.001\}$). Principals choose a channel using satisfaction scores that include unfilled requests, fees, and search costs, whereas the output gap compares realized output among completed matches. Outsourcing is therefore not itself evidence that the broker always produces better matches.

The broker’s assessment advantage appears more consistently in ranking candidates than in predicting expected output levels. Among randomly sampled potential partners (Appendix A §8), its rank correlation is 0.95 at the baseline and has a median of 0.94 across regimes, compared with 0.63 and 0.64 for principals. The broker-minus-principal difference in rank correlation is positive in 80 of 80 regimes, with paired 95% intervals above zero in 80 regimes. Ranking determines which candidates are considered first, while predicted levels also determine whether offers clear the reservation threshold.

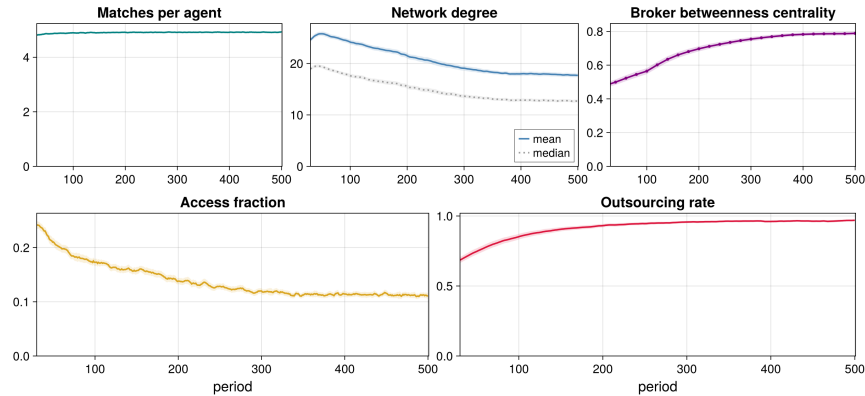


Figure 1: Baseline dynamics. Lines are 5-observation rolling ensemble means over seeds at the baseline regime; shaded ribbons are pointwise 95% Monte Carlo intervals. Some intervals are narrower than the plotted lines and therefore are not separately visible. Broker betweenness centrality is measured every 20 periods; the other outcomes are measured every period. Axes start at $t = 30$ to omit the initialization transient.

Table 1: Broker use, access, and assessment quality. Baseline columns report early- and late-window ensemble means; the across-regime column reports the median late-window mean across the 80 regimes. Outsourcing rate is outsourced requested positions divided by all requested positions.

Measure	Baseline		Across regimes
	early	late	late, median
<i>Market use and access</i>			
Outsourcing rate	0.78	0.97	0.96
Output gap (broker – self)	−0.39	+1.71	+1.54
Access fraction	0.20	0.11	0.11
<i>Assessment quality</i>			
Broker rank correlation	0.90	0.95	0.94
Principal rank correlation	0.68	0.63	0.64

Accurate ranking is therefore more important than accurate prediction of expected output levels, although both can affect matching.

Most brokered placements therefore reuse ties that principals could search directly; relatively few expand the principal network. The broker’s ranking advantage appears under pure general quality as well as complementarity: at $\rho = 1$, the broker-minus-principal difference in rank correlation averages 0.26 and is positive in all 8 pure-quality regimes. The OAT checks also show that initial network degree (k_G), broker roster fraction, and the number of strangers available in self-search have comparatively small effects on the main economic outcomes.

C The Broker’s Information Advantage Comes from Data Volume and Relational Information

The theory identifies two sources of the broker’s information advantage: the greater volume of observations generated by working across principals and relational information about both parties to each match. The Ridge experiments separate these sources from the flexibility of the neural network. When Ridge regression replaces both neural networks, the broker-minus-principal difference in rank

correlation is 0.374 at the baseline, has a median of 0.354 across the 80 regimes, and is positive in all 80. The full Ridge model still gives the broker more observations and more information about each pair than a principal. Because each restriction changes later matching and learning, their effects need not be additive.

Data volume accounts for much of the advantage (Fig. 2). Across the 31 regimes in the baseline and $\rho \times \delta$ design, the median broker-minus-principal difference in rank correlation is 0.440 in the full Ridge model but only 0.056 when the broker trains on the same number of observations as a typical principal. The difference remains positive in 27 of 31 regimes under this restriction, so data volume is important without being the broker’s only advantage.

The content of the broker’s data also matters. When each observation retains the characteristics of only one party to a match, the broker’s median advantage over principals in ranking new candidates is -0.253 and remains positive in only 8 of the 31 tested regimes. Retaining both parties’ characteristics improves on this one-party version in 30 of 31 regimes, and retaining how those characteristics interact improves it further in 30 of 31 regimes. Interaction information matters most when complementarity contributes strongly to expected match output and not at all under pure general quality. The broker’s advantage therefore comes from both the volume of observations generated through repeated vicarious relational work and information about how the characteristics of two parties combine.

D The Nature of the Matching Problem Shapes the Broker’s Advantage

The model separates two features of assessment: whether expected match output depends on general quality or pair-specific complementarity, and how difficult complementarity is to learn. The parameter ρ moves expected match output from pure complementarity at $\rho = 0$ toward pure general quality at $\rho = 1$. The parameter δ increases difficulty by making expected output change more sharply across a boundary in type-pair space. Because complementarity disappears at $\rho = 1$, δ has no effect there. The pure-quality case is therefore a separate boundary condition, not a smooth continuation of the interior ρ values.

Broker centrality is governed primarily by the composition of expected match output. In the one-at-a-time sweep from $\rho = 0$ to $\rho = 0.85$, the access fraction rises from 0.10 to 0.14 and broker betweenness centrality falls from 0.86 to 0.51. The $\rho \times \eta$ grid shows the same directions at every turnover rate. At the pure-quality boundary, centrality falls further to 0.15, while access returns to 0.11. The boundary therefore extends the decline in centrality but not the increase in access below $\rho = 1$.

Under pure complementarity, greater difficulty lowers the ranking accuracy of both the broker and principals, but it does not eliminate the broker’s advantage. Increasing δ from 0 to 1 lowers the broker’s rank correlation from 0.92 to 0.69 and the principal correlation from 0.71 to 0.46 (Fig. 3). The corresponding broker-minus-principal differences are 0.21 and 0.23. The size of this difference is not monotonic in difficulty, but its paired 95% interval is above zero in 31 of the 31 regimes. Because δ changes only complementarity, it is inactive at the pure-quality boundary.

The broker’s realized-output advantage is greatest when expected match output depends more heavily on complementarity. Paired 95% intervals for the broker-minus-self-search output gap are above zero in 31 of the 31 regimes. Composition and difficulty therefore shape different outcomes. Greater difficulty makes candidates harder for both sides to rank but does not produce a monotonic change in the broker’s relative ranking advantage. Greater reliance on complementarity increases the mean realized output produced through brokered matches. The broker retains both a ranking advantage and an output advantage throughout the grid.

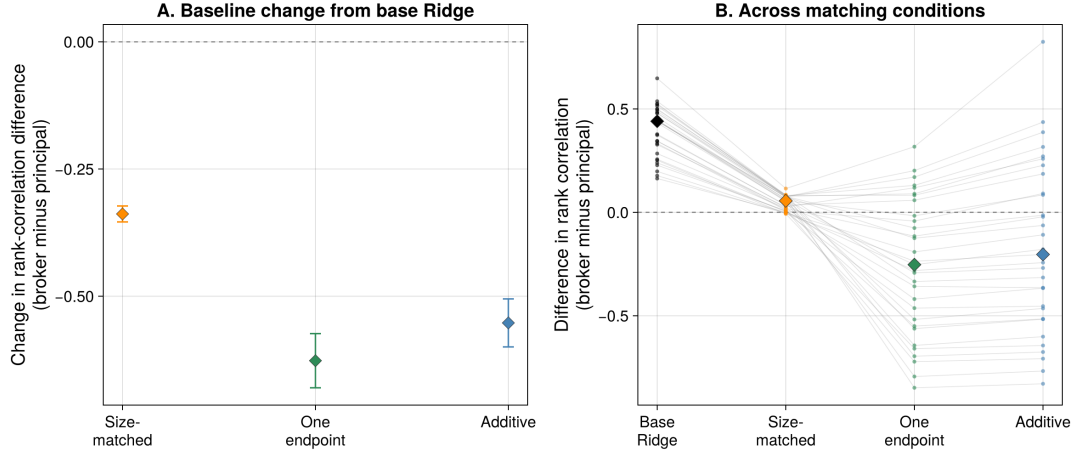


Figure 2: Sources of the broker's ranking advantage. Panel A shows each Ridge restriction's change in the broker-minus-principal difference in rank correlation at the baseline, calculated as the restriction minus the unrestricted Ridge specification. Negative values indicate that the restriction reduces the broker's ranking advantage. Points are mean differences calculated within matched runs using the same random seeds; whiskers are paired 95% Monte Carlo intervals over 50 seeds. Panel B shows the broker-minus-principal difference in rank correlation under the unrestricted Ridge specification and each restriction across the 31 regimes in the $\rho \times \delta$ design, including the baseline. Thin lines connect estimates for the same regime across Ridge variants, small points are regime means, diamonds are medians, and the horizontal line marks zero.

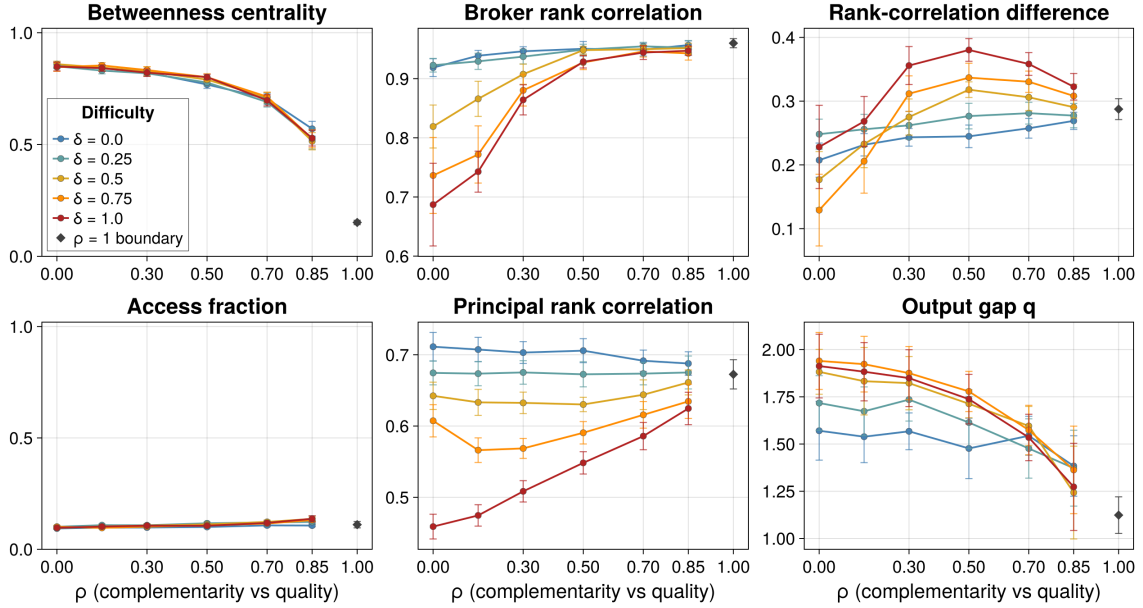


Figure 3: Outcomes across the matching problem ($\rho \times \delta$). Six outcomes across the $\rho \times \delta$ grid; one line per difficulty level δ , each spanning $\rho = 0$ through 0.85. The pure-quality boundary $\rho = 1$, where δ has no effect, is one separate regime shown once and unconnected. The figure therefore displays all 31 regimes in the design. Each marker is a late-window mean over that regime's seeds; whiskers are 95% Monte Carlo intervals. The reported difference in rank correlation is broker minus principal; the output gap is mean realized match output through the broker minus mean realized match output through self-search.

E A Broker Can Be Highly Central While Doing Little Bridging

In most regimes, the broker’s structural position strengthens over time while principals have fewer direct ties on average. At the baseline, mean principal degree falls from 25.4 to 17.7, while broker betweenness centrality rises from 0.55 to 0.79 (early to late means). This combination occurs in 64 of 80 regimes, and principal degree falls in 64 of them (Fig. 4). In most regimes, direct ties do not accumulate fast enough to offset turnover. The broker’s structural advantage therefore does not disappear as brokered matches create direct ties. Because outsourcing connects current clients to the broker, sustained use converts the value of assessment into structural centrality.

The distinction between assessment and access helps explain this relationship. An access match creates a direct tie between previously unconnected principals and can remove the broker from their shortest path. An assessment match reuses an existing tie, so the broker can serve the market without directly adding ties to the principal network. The two controlled comparisons in Figure 4 separate matching composition from turnover. Holding turnover fixed, moving from pure complementarity toward $\rho = 0.85$ raises access and lowers centrality. Holding ρ fixed, increasing turnover raises both access and centrality at every tested ρ . Turnover creates network gaps, while matching composition determines how often brokerage closes them with direct ties.

The turnover effect is substantial but nonlinear. Holding other parameters at baseline, increasing η from 0 to 0.03 lowers late mean principal degree from 77.4 to 16.3 and raises broker centrality from 0.13 to 0.82, with most of the change occurring between $\eta = 0.001$ and 0.01. Turnover also changes the broker’s realized-output advantage. In the $\rho \times \eta$ grid, the 95% interval for the broker-minus-self-search output gap is below zero in 5 of 5 ρ conditions at $\eta = 0$, but above zero in 5 of 5 by $\eta = 0.01$. In the model, turnover removes principals and ties, creating network gaps and the conditions under which brokered matches outperform self-search.

The pure-quality boundary confirms that low access alone is not sufficient for high centrality. Although higher turnover raises centrality there, centrality remains below pure complementarity at the same turnover rate. In the tested regimes, high centrality occurs when turnover renews network gaps and brokerage mainly assesses existing relationships.

Figure 5 shows that structural position relates differently to relative ranking and realized match output. Holding all other parameters fixed, moving from pure complementarity to pure general quality reduces the output gap from +1.27 to +0.54, while increasing the broker-minus-principal difference in rank correlation from 0.18 to 0.26. The broker is therefore most central and produces its largest realized-output advantage where its advantage over principals in ranking candidates is smaller. Taken together, the results separate the sources of informational, economic, and structural advantage. Observing matches across clients, while retaining information about both parties, improves candidate ranking. Complementarity increases the realized output produced by brokered matches. Turnover repeatedly removes ties, renewing the network gaps that support broker centrality. These advantages need not move together.

F Propositions

The simulation results motivate the following falsifiable propositions for future empirical research. These propositions extend the model’s implications beyond the simulated setting:

P1. In markets where match quality depends on complementarity, brokers add value primarily by assessing reachable counterparties rather than by bridging disconnected parties. The value of this assessment role increases as complementarity contributes more to match quality.

P2. A broker’s betweenness centrality is produced by sustained outsourcing, which depends on client satisfaction. It is greater where match quality depends more heavily on complementarity, and assessment-heavy brokers tend to retain or gain centrality as networks evolve.

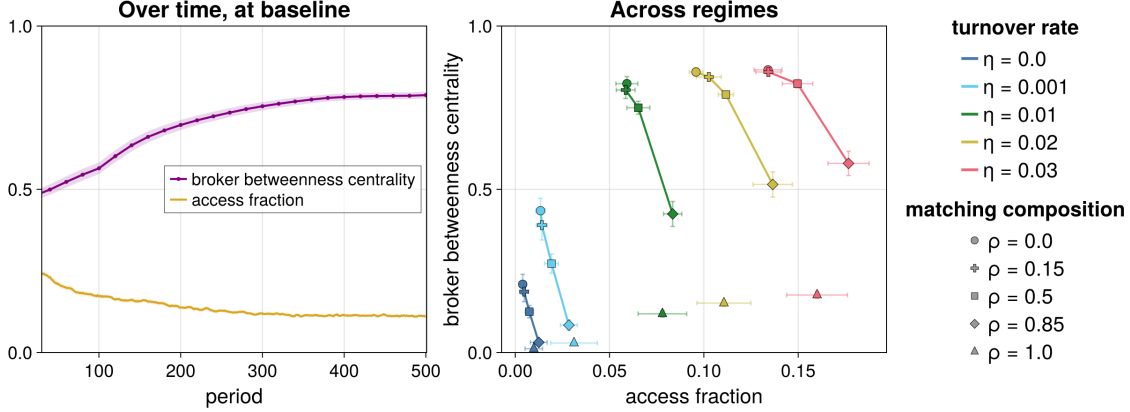


Figure 4: Broker centrality and access fraction. Left: broker betweenness centrality and access fraction over time at the baseline, as 5-observation rolling ensemble means over seeds, with pointwise 95% Monte Carlo intervals; centrality is measured every 20 periods. Right: all 25 combinations in the $\rho \times \eta$ grid, with other parameters held at baseline, including $\delta = 0.5$, each shown as a late-window mean over that regime’s seeds. Color identifies turnover η , marker shape identifies ρ , and whiskers are 95% Monte Carlo intervals on both axes. Within each η , lines connect all ρ values below 1. The pure-quality boundary $\rho = 1$, where complementarity is absent, is shown unconnected. Time-series panels begin at $t = 30$ to omit the initialization transient.

P3. Holding market turnover constant, bridging position and bridging behavior are negatively related: assessment-heavy brokers tend to be more central while doing less bridging. Higher turnover can raise both access and centrality by continually creating new network gaps.

P4. Brokers can provide superior assessment by ranking potential matches better than principals even when their predictions of expected output levels are inaccurate.

These propositions explain how brokers accumulate and sustain informational and structural advantages. The next section considers how a corporate broker may convert accumulated information into resource capture or data enclosure.

V Transient Brokerage and Its Conditions of Possibility

The model formalizes the accumulation of informational and structural advantage, but it stops before the transition to capture. I now return to the theoretical argument to explain how a corporate broker may convert those accumulated advantages into ownership or control.

Outsourced relational work performed at scale by corporate brokers creates the conditions under which this conversion can occur. The broker dismantles the relational package previously negotiated with its clients and replaces it with a new configuration organized around the broker’s ownership of the resources it used to intermediate (*resource capture*), or the monetization of the information generated through prior matching activity (*data enclosure*).

These configurations are two qualitatively different forms of capture, which can be pursued in parallel and along with standard brokering. In *resource capture*, the broker takes inventory risk and starts acting as a principal in the market it used to intermediate. Counterparties interact only with the broker, without creating direct ties with one another. As the broker expands its principal activity and shrinks its matching services, former clients become counterparties or competitors. In *data enclosure*,

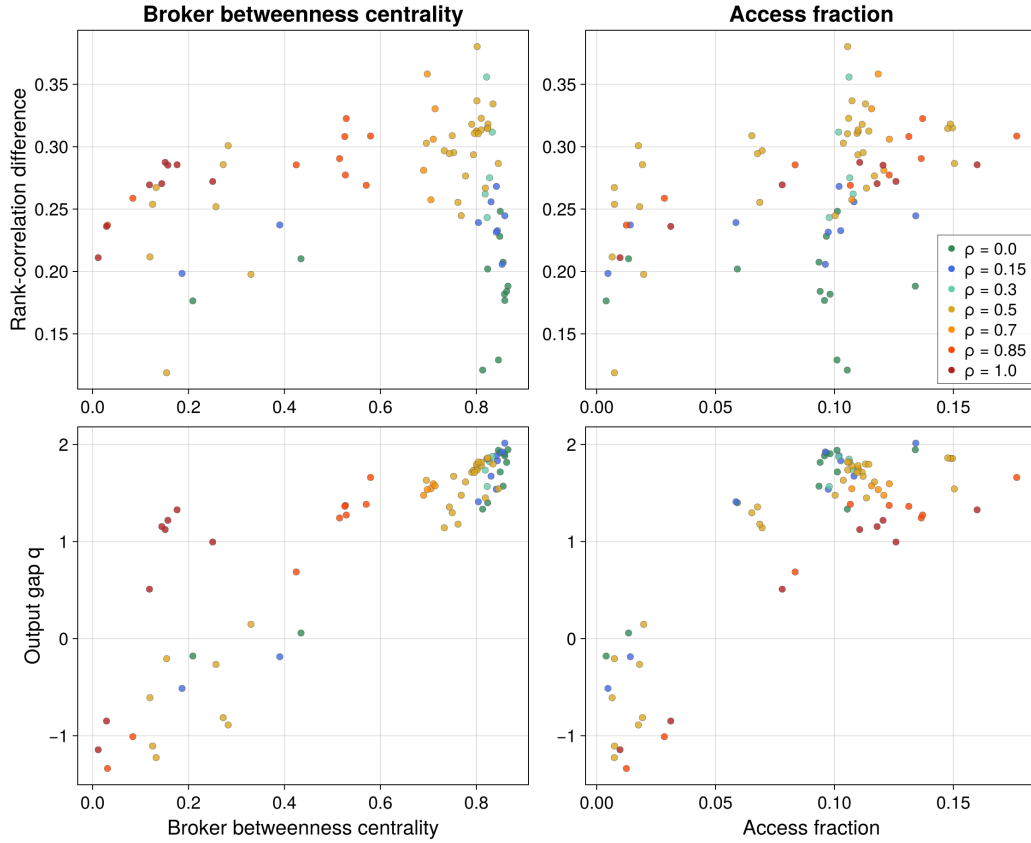


Figure 5: Ranking and output differences by centrality and access. Difference in rank correlation (broker minus principal) and output gap (mean realized match output through the broker minus mean realized match output through self-search) against broker betweenness centrality and access fraction; one dot per regime, each a late-window mean over its seeds. Color identifies matching composition ρ .

the broker monetizes information directly by selling predictions, analytics, or other information services, often as a subscription.

What makes capture possible, I suggest, is primarily the broker’s information advantage, not its structural position. Structural and informational advantages are decoupled, and capture may occur even if the broker’s centrality is declining. This is because constructing viable relational packages generates an informational byproduct that the broker can leverage. Two features of the broker’s informational landscape facilitate capture: cumulative knowledge of complementarities coming from the *outsourced* nature of its relational work, and information infrastructures arising from its *corporate* nature. Other actors cannot easily replicate either advantage.

(i) *The role of complementarity information.* The broker’s vicarious work generates knowledge of how different parties pair together, what kinds of matches succeed across the market, and what relational packages support different transactions. For regular market actors, this complementarity information is hard to replicate. Unlike attributional information about the general quality of counterparties, which any actor with sufficient market exposure can in principle acquire, complementarity information is generated through the repeated observation of various pairings. The broker holds it because it does the matching for a variety of clients; clients, who experience only their own matches, do not—unless they purchase or otherwise acquire it. Complementarity information supports resource capture, because it allows the broker to identify which resources or counterparties are most likely to generate profitable matches, so that it can capture them strategically and offload them gainfully. Complementarity information also supports data enclosure because it can be packaged into prediction or analytics products and services.

(ii) *The role of information infrastructures.* The corporate broker’s information infrastructure explains *how* its information advantage can be preserved and converted. The taxonomies, classification systems, contractual templates, and algorithmic sorting procedures that the broker develops to perform vicarious work at scale and in a distributed manner, through the everyday work of its employees and representatives, are also the infrastructures through which capture can be realized. They can be repurposed to route resources to the broker itself, to put these resources back in circulation profitably, or to organize data into a marketable product. Moreover, the corporate form gives the broker the legal privileges required to build, own, and protect these infrastructures over time (Coleman 1982; Pistor 2019).

These features, and transient brokerage itself, can be found across markets. I illustrate each configuration with an example.

Dealers in collectible markets. Collectors of art, fine wine, rare books, or similar specialty goods may seek trades through broker-dealers who know the market. Each collector has distinct tastes and holdings; match quality depends on multidimensional complementarity between what one party has and what another wants. Dealers accumulate knowledge of collector preferences across transactions, and they may decide to leverage it and start holding inventory, for instance, by launching and operating a gallery.

Interdealer brokerage in over-the-counter (OTC) financial markets. Interdealer brokers (IDBs) are specialized intermediaries in OTC financial markets (e.g., bond markets, foreign exchange, commodities). IDBs help institutions like banks and dealers conclude large, complicated, or anonymous transactions in the absence of a centralized trading place. They solve a hard matching problem: the right counterparty for a trade depends on inventory positions and risk appetites that parties wish to keep secret. IDBs must know who wants what, negotiate prices across hidden order interest, preserve anonymity until execution, and manage long-term client relationships to remain in the order flow. The complementarity information they accumulate, about who will absorb what positions at what price under what conditions, is substantial. Resource capture, however, faces regulatory and reputational hurdles. Instead, IDBs have monetized accumulated information through data divisions

that package historical OTC pricing and other data products and sell them to the same dealers whose trades generated the data and to a wider set of buy-side firms and regulators (see, e.g. Parameta Solutions 2026; TP ICAP 2026).

In summary, three elements enable capture or enclosure: a matching problem that makes complementarity information valuable, an informational advantage accumulated through repeated vicarious relational work, and an institutional architecture that allows that information to be aggregated, organized, and protected.

VI Discussion

A Structural Advantage as a Product of Relational Work

Within the agent-based model, the broker’s centrality is a product of its relational work. The broker is central because many principals outsource to it repeatedly, which they do because its assessments are good. Its centrality rises and falls with outsourcing and satisfaction, not with exogenous features of the initial graph.

The simulations show that the broker’s relational work endogenously produces its structural advantage. This process generates two patterns that existing theory does not readily predict. The first is that the broker tends to be most central exactly where it bridges least in practice. Across variation in matching composition at a fixed turnover rate, centrality is negatively associated with the access fraction (Figure 4): in the complementarity-dominated markets where the broker is most prominent, almost all its matches re-pair principals already tied to one another. A strong bridging position coincides with weak bridging behavior. Occupying a strategic position and using it for one’s work are different things that theory should not conflate.

The second pattern is about the sustainability of brokerage. A long-standing expectation in the tradition is that brokerage tends to be self-undermining (Burt 1992, 2002; Stovel, Golub, and Milgrom 2011). Each successful introduction densifies the network and dissolves the structural hole that gave the broker its advantage. Where the work is assessment, however, it leaves the network no denser than it found it. Across the simulations, access work is low and declining over time, so the broker’s structural advantage tends to persist. Self-liquidation is a feature of access brokerage. Meanwhile, the assessment-heavy brokerage characteristic of complementarity-based matching markets is more durable.

The results also suggest a precise mechanism for the decoupling of structural and informational advantage posited in Section V. The two advantages respond to different features of the matching problem. The broker’s structural prominence tracks the composition of match value, rising as matching becomes more complementarity-driven, while its prediction edge responds to the difficulty of learning the complementarity component, which can vary independently. A broker can be highly central despite a modest informational lead or hold a decisive informational lead from an unremarkable position. Betweenness centrality is best understood not as a static measure of opportunity but as a product of the relational work the broker performs.

B Scope Conditions and Generalizability

The concept of transient brokerage applies most directly to matching markets in which (i) matching is complex enough that the value of a match depends not only on the general quality of each party but also on their complementarity and (ii) a corporate broker operates as a repeat-player accumulating information across transactions. The further a case departs from these conditions, the weaker the predictions of the theory. When the relational dimension is weak, the broker’s advantage rests primarily on data volume about multiple counterparties, making its informational position less

privileged. When the broker is a physical person, she lacks the temporal persistence, infrastructural capacity, and legal architecture that the corporate broker relies on.

The dynamics of transient brokerage also depend on the nature of the brokered resource and the institutional environment. In a goods market where items are bulky and physical delivery is required, the costs and logistics of capture differ substantially from those in a market for services or a market where goods are virtual or ownership passes without physical possession. In some markets, regulation may slow or prevent capture. Market structure matters as well. The model considered in this article features a single broker. With multiple brokers competing for the same clients or matches, the cross-market data pool is fragmented, and each broker may accumulate a smaller informational advantage, unless one or a few brokers dominate the market. A systematic treatment of broker competition is deferred to future work.

Transient brokerage is relevant to a particular configuration of brokerage, in which the matching problem, corporate form, and information accumulation give rise to a distinctive trajectory. The sociological literature has used brokerage to describe a wide range of phenomena, from immigrant children translating for their parents to marriage matchmakers and political brokers. The relational-work view of brokerage developed in this article can be applied to what many of these so-called brokers do. Without the repeat-player structure of corporate brokerage, the legal and infrastructural architecture of the firm, and an economic matching problem complex enough to sustain an informational advantage, however, the pathway to capture is not the same.

These scope conditions point to candidate domains in which to test the theoretical propositions. Real estate and various types of financial intermediation, including interdealer brokerage, combine complex matching, corporate repeat-player brokers, and meaningful complementarity information. Import-export management and data markets also share many of these features.

C The Political Economy of Corporate Intermediation

The corporate broker is a vehicle for the accumulation of information generated by the interactions between others, and the principals whose matching activity produced the information do not participate in the profits. The asymmetries that make capture possible are not just a byproduct of a broker's size or expertise. They are legally constituted. Corporate brokers assemble cross-market data and build classification systems and information infrastructures under the legal protections of the corporate form, which allows vicarious relational work to solidify into a durable informational moat. Whether a broker can convert its informational position into resource ownership or data provision depends on the institutional environment as much as on network position.

When the broker becomes a principal, it changes who gets connected to whom. The principals observe far fewer matches than they otherwise would, and most matches in the market teach them nothing at all. The informational asymmetry that makes capture possible is widened by capture itself: the broker takes control not only of the resource but of the data its placement generates. Transient brokerage shifts how value is distributed, channeling the returns of participants' collective relational activity to the corporations that record and mine the data it generates.

Capture by corporate brokers has become more common since the 1980s alongside three transformations that have reshaped market intermediation: the turn to finance, which expanded the share of economic activity routed through specialized intermediaries (Carruthers 2015; Krippner 2011); the digital turn, which dramatically lowered the cost of accumulating, storing, and analyzing the data that brokerage generates while also expanding the demand for predictive analytics (Cohen 2019; Gandy 1993); and the rise of platform capitalism, which produced multi-sided market operators that own the relational infrastructure of matching while monetizing the data it generates (Sadowski 2020; Viljoen, Goldenfein, and McGuigan 2021). Transient brokerage is one of the mechanisms that link

these transformations together and contributes to a broader shift in how value is extracted from economic activity.

Significant evidence from financial markets shows that corporate brokers that once facilitated matches between others have increasingly become principals, either by capturing the activity they used to intermediate or by monetizing data. The demutualization of stock exchanges is a prime example. As late as the mid-1990s, almost all major global exchanges operated as member-owned mutual organizations. By the mid-2000s, a majority had introduced electronic trading systems that allowed them to grant trading access directly to investors, bypassing the services of member firms, and had converted to for-profit shareholder structures (Aggarwal 2002; Petry 2021). For-profit exchanges now derive a growing share of revenue from proprietary data feeds and trading technology (MacKenzie 2021; Pardo-Guerra 2019).

A parallel transformation has taken place in interdealer brokerage for over-the-counter (OTC) markets, where the data divisions of prominent brokers now generate substantially higher operating margins than the broking activities themselves; this is the case, for instance, of the Parameta Solutions data division of TP ICAP, the largest interdealer broker (TP ICAP 2026). In consumer markets, Visa offers another example. Reorganized in 1970 as a member-owned cooperative of issuing banks, Visa operated for decades as a non-profit utility that brokered payment authorizations between cardholders, merchants, and banks (Stearns 2011). Following its 2007 restructuring and 2008 IPO, Visa converted into a for-profit corporation that now derives substantial revenue from data, analytics, and risk solutions services, alongside the network and processing fees generated by its core authorization and clearing activities (Visa Inc. 2024).

VII Conclusion

This article reconceptualizes brokerage as outsourced relational work and explains how repeated matching can produce an informational advantage that a corporate broker may later convert into resource capture or data enclosure. The formal illustration demonstrates one possible pathway through which repeated vicarious relational work can produce informational and structural advantages, and it generates propositions about when those advantages should arise.

The very conditions that make brokerage possible and valuable, such as the scarcity of information and the complexity of the matching problem, create opportunities for the broker—particularly the corporate broker—to take control of resources. At the transition point, the broker dismantles the relational package negotiated with its principals and collapses the boundary between *vicarious* and *personal* relational work to pursue its own advantage. It abandons the service-tie configuration that existed and (re)negotiates a new one, premised on the broker’s direct ownership and control of formerly brokered resources or the data they generated. In the latter case, the accumulated data can also be mined internally or licensed out to feed proprietary behavioral prediction, classification, and scoring products (Fourcade and Healy 2017, 2024) for use in markets distinct from those in which they originated (Viljoen 2021; Zuboff 2019).

This article helps explain why the digital turn has facilitated broker capture at scale. Because information is central to brokerage, changes in the information ecosystem reshape the conditions under which transient brokerage becomes available, and digital technology has changed those conditions decisively. Relational infrastructures that once consisted of internal handbooks, contractual templates, and bureaucratic procedures now incorporate data taxonomies, algorithmic sorting, and interface designs that constitute the relational packages they administer in real time and at scale. Digital technology amplifies these asymmetries and shortens the path from accumulated information to ownership or monetization.

A Digital Platforms and Automated Matching

Transient brokerage is intimately related to the rise of digital platforms and algorithmic marketplaces, from gig economy apps to electronic exchanges, that partly automate the work of matching. Where brokers perform vicarious relational work on behalf of their clients, platform operators design mechanisms to automate and circumvent relational work, typically through auction protocols, ranking algorithms, and rating systems. The platform operator does not directly match buyers and sellers; it builds the apparatus that matches them and extracts fees or data flows from the result.

In some markets, mechanism design substitutes for relational work. In others, it absorbs and reshapes relational work, with the platform operator administering the categories, ratings, and defaults that constitute the relational packages on behalf of both sides. The dynamics of their emergence share important features with transient brokerage: the platform operator accumulates cross-market information about its users, holds infrastructural and legal advantages that make this accumulation difficult to challenge, and is often well positioned to transition from automating matches to selling matched resources or analytics about them.

For example, Amazon operates as both a marketplace for third-party sellers and a direct retailer of its own products. Zhu and Liu (2018) tracked over 150,000 product listings initially supplied by third-party sellers and found that Amazon began directly selling roughly three percent of them over the following ten months. The platform was significantly more likely to enter listings with higher sales and better customer reviews, using information available to it as the marketplace operator. Khan (2017, 2019) argues that platform operators in this position are best understood not as intermediaries but as integrated firms whose control of the matching infrastructure gives them a structural advantage over the sellers they host. The same configuration recurs beyond Amazon, across leading platform firms: their business model becomes that of a lean platform that owns and profits from the matching infrastructure while pushing the underlying productive activity and capital risk onto third parties (Rahman and Thelen 2019).

B Future Directions

Two main directions warrant future work. On the modeling side, the ABM developed here is exploratory. It has many interacting components and considerable researcher degrees of freedom in its specification, and it carries the usual caveats of this kind of hand-crafted simulation: its results illustrate the theory rather than establish it. Systematic investigation of the parameter space, the decision rules, and the model's sensitivity to them may surface dynamics that the results presented in this article do not reveal. The resource capture and data enclosure mechanisms are introduced theoretically but not formalized; they are natural extensions. On the theoretical side, the framework does not yet fully address how platforms and algorithmic marketplaces reshape the dynamics of brokerage by automating the vicarious relational work that brokers historically performed. Both directions point toward the same underlying question about the impact of the digital turn on brokerage, the institutions that rely on it, and the actors that perform it or benefit from it.

The digital turn has brought about a series of transformations. The informational byproduct of relational matching has come to rival the value of the matches themselves. Relational work is increasingly valued for the data it generates and performed by machines rather than people. The informational asymmetry between brokers and their principals has widened. Existing relational packages are being dismantled to extract value from the information and infrastructures associated with them. These shifts redraw the boundaries on which brokerage rests: between internalized and outsourced relational work, and between vicarious and personal relational work. Brokerage becomes a transient position, held so long as brokering remains more profitable than capture.

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Appendix A

Simulation Pseudocode for an ABM of Brokerage in Matching Markets

1 State, Notation, and Matching Function

Blue notation, equations, and algorithm lines are diagnostics-only measures. They are recorded for $\mathcal{M}$ or downstream figures, exploration scripts, and analyses; they do not enter transition, choice, learning, or matching rules.

For any matrix Z , Z_ℓ denotes column ℓ and $Z_{.L}$ denotes the submatrix with ordered column-index set L . For a vector z , z_L denotes the corresponding subvector.

Table 1: Cross-routine symbols and dimensions.

Symbol	Dimension/domain	Definition
<i>Core</i>		
N	scalar	Number of agents, with agent set $[N] = \{1, \dots, N\}$.
ϑ	parameter vector	Simulation parameter vector.
seed	integer	Random seed.
S^t	state tuple	Mutable simulation state at end of period t .
T	integer	Number of simulation periods.
$\mathcal{M} = [m^1; \dots; m^T]$	$\mathbb{R}^{T \times m}$	Measure table; row $m^t = \mathcal{M}_t \in \mathbb{R}^m$.
<i>Parameters</i>		
d	scalar	Agent type dimension.
k_G	scalar	Watts-Strogatz graph degree.
ω	scalar	Satisfaction EWMA weight.
$\mathbf{L}$	learning function	NN (default) or Ridge.
$\mathbf{S}_b$	broker service	Full (default), assessment-only, or access-only.
λ_a, λ_b	positive scalars	Agent and broker Ridge slope penalties; calibrated experiment values are $\lambda_a = 0.003$ and $\lambda_b = 0.001$.
a_b	broker Ridge variant	Pair, size-matched, single-principal, or additive.
$E_{\text{init},a}, E_{\text{init},b}$	scalars	Agent and broker initial training steps; both are 100 in the reporting simulations.
$E_{\text{steps},a}, E_{\text{steps},b}$	scalars	Agent and broker minimum training steps per fitting call; both are 50 in the reporting simulations.
$\eta_{\text{r},a}, \eta_{\text{r},b}$	scalars	Agent and broker Adam learning rates; reporting values are 0.003 and 0.03, respectively.
W_p	scalar	Training-window horizon, in periods.
M	scalar	Per-call training observation cap.
<i>Agent types, networks, and output</i>		
$G^t = (V, \mathcal{E}_G^t)$	graph	Undirected graph, $V = [N] \cup \{N + 1\}$; node $N + 1$ is the broker.

Symbol	Dimension/domain	Definition
$X^t = [x_1^t, \dots, x_N^t]$	$\mathbb{R}^{d \times N}$	Agent type matrix; $x_i^t = X_{\cdot i}^t \in \mathbb{R}^d$.
γ	$[0, 1] \rightarrow \mathbb{R}^d$	Agent type curve for initial and entrant draws.
q_{ij}^t	scalar	Realized output for match (i, j) .
<i>Match-output function</i>		
Q	scalar	Output offset.
ρ	scalar	General-quality share of the two components' summed marginal variances.
δ	scalar	Strength of the pair boundary in match-specific complementarity.
σ_ε	scalar	Output-noise standard deviation.
c	$\mathbb{R}^d$	Ideal agent type.
A	$\mathbb{R}^{d \times d}$	Symmetric positive definite complementarity matrix.
B	$\mathbb{R}^{d \times d}$	Symmetric boundary matrix.
<i>Calibration and costs</i>		
$\bar{q}_{\text{cal}}$	scalar	Calibration mean.
r	scalar	Reservation output.
ϕ	scalar	Broker fee.
c_s	scalar	Self-search cost.
<i>Learning and predictions</i>		
d_b	integer	Broker symmetric pair-feature dimension, $d + d(d + 1)/2$.
$\psi_b(x_i, x_j)$	$\mathbb{R}^{d_b}$	Symmetric broker pair feature vector.
$\mathcal{H}_i^t = (\Xi_i^t, y_i^t, n_i^t)$	$\Xi_i^t \in \mathbb{R}^{d \times n_i^t}, y_i^t \in \mathbb{R}^{n_i^t}$	Agent i history; observation ℓ is partner agent type $\Xi_{i,\ell}^t$ and output $y_{i\ell}^t$.
$\mathcal{H}_b^t = (X_{b,1}^t, X_{b,2}^t, y_b^t, n_b^t)$	$X_{b,1}^t, X_{b,2}^t \in \mathbb{R}^{d \times n_b^t}, y_b^t \in \mathbb{R}^{n_b^t}$	Broker history; observation ℓ stores the two party types and output $y_{b\ell}^t$.
Θ_i^t	predictor parameters	Agent i NN weights or Ridge coefficients.
Θ_b^t	predictor parameters	Broker NN weights or Ridge coefficients.
$\mathcal{O}_i^t, \mathcal{O}_b^t$	Adam states	Persistent NN optimizer moments and step counters; unused by Ridge.
Δ_i^t, Δ_b^t	nonnegative integers	Observations added since the learner's previous training call.
$\mathbf{p}_i^t, \mathbf{p}_b^t$	integer sequences	Cumulative history counts at the end of learner initialization (period 0) and each later completed period.
$v_b^t(i, j)$	scalar	Symmetric broker prediction for unordered pair $\{i, j\}$.
$a_{i \rightarrow j}^t$	scalar	Agent i 's assessment of candidate j .
$w_{j \leftarrow i}^t$	scalar	Receiver acceptance score.
$\bar{q}_i^t$	vector	Agent i 's partner-indexed empirical means.
C_i^t	vector	Agent i 's partner-indexed counts.
<i>Period state and market objects</i>		
s_i^t	$\mathbb{R}^2$	Satisfaction scores for self and broker channels.
χ_i^t	binary	Indicator that agent i has ever used the broker.
M_i^t	set	Agent i current-period counterparties.
z_i^t	channel	Channel chosen by demander i .
d_i^t	integer	Current-period demand.
D^t	subset of $[N]$	Current broker clients, $\{i : d_i^t > 0, z_i^t = \text{broker}\}$.
$\mathcal{A}_b^t$	subset of $[N]$	Broker access set, $\text{Roster}^t \cup D^t$.
Roster^t	subset of $[N]$	Standing roster.

Symbol	Dimension/domain	Definition
rep_b^t	scalar	Broker reputation.
D_+^t	subset of $[N]$	Broker clients with positive residual offer quota.
$\mathcal{D}^t$	list	Demand records (i, z_i^t, d_i^t) with $d_i^t > 0$.
$\mathcal{R}^t$	list	Accepted records $(i, j, z, q, \hat{q}, o_1, o_2)$.
rng^t	RNG state	State of the simulation random-number stream.

The mutable simulation state at the end of period t is

$$S^t = \left(\vartheta, \text{rng}^t, \gamma, c, A, B, \bar{q}_{\text{cal}}, r, \phi, c_s, X^t, G^t, \right. \\ \left. \{\mathcal{H}_i^t, \Theta_i^t, \mathcal{O}_i^t, \Delta_i^t, \mathbf{p}_i^t, \bar{q}_i^t, C_i^t, s_i^t, \chi_i^t, M_i^t\}_{i=1}^N, \right. \\ \left. \mathcal{H}_b^t, \Theta_b^t, \mathcal{O}_b^t, \Delta_b^t, \mathbf{p}_b^t, \text{Roster}^t, D^t, \text{rep}_b^t \right).$$

This tuple includes fixed realized model objects and all variables needed for later transitions. Implementation-only scratch buffers and diagnostic-only counters are omitted.

For the fixed reference moments, define

$$U_{ij} = \frac{x_i^\top c + x_j^\top c}{2}, \quad V_{ij} = x_i^\top A x_j, \quad H_{ij} = \text{sign}(x_i^\top B x_j) V_{ij}, \\ \tilde{U}_{ij} = \frac{U_{ij} - \mu_U}{\sigma_U}, \quad \tilde{W}_{ij}(\delta) = \frac{V_{ij} + \delta H_{ij} - (\mu_V + \delta \mu_H)}{\sqrt{\sigma_V^2 + 2\delta \text{Cov}(V, H) + \delta^2 \sigma_H^2}}.$$

With $\kappa(\delta) = \text{Corr}(U, W(\delta))$, the systematic output component is

$$f(x_i, x_j) = \frac{\sqrt{\rho} \tilde{U}_{ij} + \sqrt{1-\rho} \tilde{W}_{ij}(\delta)}{\sqrt{1 + 2\sqrt{\rho(1-\rho)} \kappa(\delta)}}.$$

Expected match output is $Q + f(x_i, x_j)$. A completed match produces realized output

$$q_{ij} = Q + f(x_i, x_j) + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_\varepsilon^2).$$

2 Top-Level Simulation

Algorithm 1 Brokerage Simulation

Require: Parameters ϑ and seed.

Ensure: Final state S^T and **measure table** $\mathcal{M} \in \mathbb{R}^{T \times m}$.

- 1: $S^0 \leftarrow \text{INITIALIZE}(\vartheta, \text{seed})$
 - 2: **for** $t = 1, \dots, T$ **do**
 - 3: $(S^t, \mathcal{R}^t, Y^t) \leftarrow \text{PERIODUPDATE}(S^{t-1}, t, \vartheta)$
 - 4: $\mathcal{M}_t \leftarrow Y^t$
 - 5: **return** $(S^T, \mathcal{M})$
-

3 Learning and Prediction

Let $\mathbf{L} \in \{\text{NN}, \text{Ridge}\}$ select the learning function. The default is $\mathbf{L} = \text{NN}$. Both branches use the same histories, period window, and observation cap. Inputs remain on their raw DGP scale.

For any one-hidden-layer predictor with input dimension p_{in} and hidden width h ,

$$\text{NN}_{\Theta}(\xi) = w_2^\top \text{ReLU}(W_1 \xi + b_1) + b_2, \quad \Theta = (W_1, b_1, w_2, b_2),$$

where $\xi \in \mathbb{R}^{p_{\text{in}}}$, $W_1 \in \mathbb{R}^{h \times p_{\text{in}}}$, $b_1, w_2 \in \mathbb{R}^h$, and $b_2 \in \mathbb{R}$.

$$d_b = d + \frac{d(d+1)}{2}, \quad \psi_b(x_i, x_j) = \left[x_i + x_j; \text{vech}\left(\frac{x_i x_j^\top + x_j x_i^\top}{2}\right) \right].$$

Agent networks use $(p_{\text{in}}, h) = (d, 2d)$; the broker network uses $(p_{\text{in}}, h) = (d_b, 8d)$.

$$\psi_b(x_i, x_j) = \psi_b(x_j, x_i)$$

so broker histories produce one training column per observed unordered pair.

Local variables for training

Symbol	Dimension/domain	Definition
p_{in}, h	integers	Predictor input dimension and hidden width.
ξ	$\mathbb{R}^{p_{\text{in}}}$	Predictor input vector.
$\mathbf{L}$	learning function	NN or Ridge.
a_b	broker Ridge variant	Pair, size-matched, single-principal, or additive.
Θ	predictor parameters	NN parameters (W_1, b_1, w_2, b_2) or Ridge parameters (α, β) .
W_1, b_1, w_2, b_2	matrix, vectors, scalar	One-hidden-layer MLP weights and biases.
d_b	integer	Broker symmetric pair-feature dimension.
ψ_b	function	Symmetric broker pair-feature map.
$\Xi_{\text{tr}}, y_{\text{tr}}$	$\mathbb{R}^{p_{\text{in}} \times n}, \mathbb{R}^n$	Generic training inputs and targets.
$\mathcal{L}$	scalar	Full-batch MSE objective.
λ	positive scalar	Learner-specific Ridge slope penalty; intercepts are not penalized.
e	integer	Update-step index.
(m, v, τ)	Adam state	Persistent Adam moments and step counter; $\mathcal{O}_i, \mathcal{O}_b$ are the agent and broker instances.
$\beta_1, \beta_2, \epsilon$	scalars	Adam constants $(0.9, 0.999, 10^{-8})$.
n_i, n_b	integers	Current values of history lengths n_i^t and n_b^t .
L_i, L_b	ordered index sets	Training-window indices from the most recent W_p completed simulation periods, with period 0 included only until W_p later periods exist; subsampled evenly to at most M .
Δ_i, Δ_b	integers	New observations since last training call.
E_i, E_b	integers	Agent and broker NN update-step counts.
$\Xi_{i,\text{tr}}, y_{i,\text{tr}}$	$\mathbb{R}^{d \times L_i }, \mathbb{R}^{ L_i }$	Agent i training view of $\mathcal{H}_i^t$.
$Z_{b,\text{tr}}, y_{b,\text{tr}}$	matrix, $\mathbb{R}^{ L_b }$	Broker fitting view using the selected input map.
r_ℓ	$\{1, 2\}$	Endpoint retained once when broker observation ℓ is recorded for the single-principal ablation.

Algorithm 2 TrainNN

Require: Network parameters Θ , inputs $\Xi_{\text{tr}} \in \mathbb{R}^{p_{\text{in}} \times n}$, targets $y_{\text{tr}} \in \mathbb{R}^n$, steps E , learning rate η_{lr} , persistent Adam state (m, v, τ) .

- 1: **for** $e = 1, \dots, E$ **do**
- 2: $g \leftarrow \nabla_{\Theta} \mathcal{L}(\Theta)$, $\mathcal{L}(\Theta) = n^{-1} \sum_{\ell=1}^n (\text{NN}_{\Theta}(\Xi_{\text{tr}, \cdot \ell}) - y_{\text{tr}, \ell})^2$
- 3: $\tau \leftarrow \tau + 1$
- 4: $m \leftarrow \beta_1 m + (1 - \beta_1)g$, $v \leftarrow \beta_2 v + (1 - \beta_2)g \odot g$
- 5: $\Theta \leftarrow \Theta - \eta_{\text{lr}} \frac{m/(1 - \beta_1^{\tau})}{\sqrt{v/(1 - \beta_2^{\tau})} + \epsilon}$
- 6: **return** $\Theta, (m, v, \tau)$

Algorithm 3 FitPredictor

Require: Learning function $\mathbf{L}$, learner $c \in \{a, b\}$, parameters Θ , raw inputs $\Xi_{\text{tr}} \in \mathbb{R}^{p_{\text{in}} \times n}$, targets $y_{\text{tr}} \in \mathbb{R}^n$, learner-specific Ridge penalty λ_c , and fitting parameters.

- 1: **if** $\mathbf{L} = \text{NN}$ **then**
- 2: $E \leftarrow \max\{E_{\text{steps}, c}, \lceil E_{\text{init}, c} \Delta / n_{\text{total}} \rceil\}$.
- 3: $(\Theta, \mathcal{O}) \leftarrow \text{TRAINNN}(\Theta, \Xi_{\text{tr}}, y_{\text{tr}}, E, \eta_{\text{lr}, c}, \mathcal{O})$.
- 4: **else**
- 5: $(\alpha, \beta) \leftarrow \arg \min_{a, b} \left\{ n^{-1} \sum_{\ell=1}^n (a + b^{\top} \Xi_{\text{tr}, \cdot \ell} - y_{\text{tr}, \ell})^2 + \lambda_c \|b\|_2^2 \right\}$.
- 6: $\Theta \leftarrow (\alpha, \beta)$.
- 7: **return** Θ and, for NN, $\mathcal{O}$.

The agent input is always the prospective partner type. The broker Ridge training input for observation ℓ is

$$z_{b\ell} = \begin{cases} \psi_b(x_{1\ell}, x_{2\ell}), & a_b \in \{\text{pair, size-matched}\}, \\ x_{r_{\ell}, \ell}, & a_b = \text{single-principal}, \\ x_{1\ell} + x_{2\ell}, & a_b = \text{additive}. \end{cases}$$

For size-matched history, uniformly sample without replacement from the broker's windowed and capped observations. Its sample size is the smaller of the broker sample size and the nearest-integer median agent sample size among agents with nonempty histories, with half-integers rounded up.

Prediction is modular:

$$\text{PREDICTAGENT}(i, x_j) = \begin{cases} \text{NN}_{\Theta_i}(x_j), & \mathbf{L} = \text{NN}, \\ \alpha_i + \beta_i^{\top} x_j, & \mathbf{L} = \text{Ridge}, \end{cases}$$

and

$$\text{PREDICTBROKER}(i, j) = \begin{cases} \text{NN}_{\Theta_b}(\psi_b(x_i, x_j)), & \mathbf{L} = \text{NN}, \\ \alpha_b + \gamma_b^{\top} \psi_b(x_i, x_j), & \mathbf{L} = \text{Ridge}, a_b \in \{\text{pair, size-matched}\}, \\ \alpha_b + \gamma_b^{\top} (x_i + x_j), & \mathbf{L} = \text{Ridge}, a_b = \text{additive}, \\ g_b(x_i) + g_b(x_j) - \bar{y}_{b, \text{tr}}, & \mathbf{L} = \text{Ridge}, a_b = \text{single-principal}, \end{cases}$$

where $g_b(x) = \alpha_b + \gamma_b^{\top} x$. Before any Ridge fit, $\alpha = Q$, slopes are zero, and $\bar{y}_{b, \text{tr}} = Q$.

Within period-level routines, state variables without a superscript refer to their current-period values.

Algorithm 4 TrainPredictors

Require: State S , period t , learning function L , and fitting parameters.

- 1: **for** $i = 1, \dots, N$ in parallel **do**
 - 2: $n_i \leftarrow n_i^t$, $\Delta_i \leftarrow$ observations added to $\mathcal{H}_i$ since its last training call.
 - 3: **if** $n_i > 0$, $\Delta_i > 0$, and $i \bmod 2 = t \bmod 2$ **then**
 - 4: Let $P_i \leftarrow |\mathbf{p}_i|$ and $b_i \leftarrow 0$ if $P_i \leq W_p$, otherwise $b_i \leftarrow (\mathbf{p}_i)_{P_i - W_p}$; set $L_i \leftarrow \{b_i + 1, \dots, n_i\}$.
 If $|L_i| > M$, keep M spaced indices across L_i , including its first and last indices (or only its last index when $M = 1$).
 - 5: $\Xi_{i,\text{tr}} \leftarrow (\Xi_i^t)_{L_i}$, $y_{i,\text{tr}} \leftarrow (y_i^t)_{L_i}$.
 - 6: $\Theta_i \leftarrow \text{FITPREDICTOR}(L, a, \Theta_i, \Xi_{i,\text{tr}}, y_{i,\text{tr}}, \lambda_a, \Delta_i, n_i, \vartheta)$; $\Delta_i \leftarrow 0$.
 - 7: $n_b \leftarrow n_b^t$, $\Delta_b \leftarrow$ broker observations since last training.
 - 8: **if** $n_b > 0$ and $\Delta_b > 0$ **then**
 - 9: Let $P_b \leftarrow |\mathbf{p}_b|$ and $b_b \leftarrow 0$ if $P_b \leq W_p$, otherwise $b_b \leftarrow (\mathbf{p}_b)_{P_b - W_p}$; set $L_b \leftarrow \{b_b + 1, \dots, n_b\}$.
 If $|L_b| > M$, keep M spaced indices across L_b , including its first and last indices (or only its last index when $M = 1$).
 - 10: If $L = \text{Ridge}$ and $a_b = \text{size-matched}$, replace L_b by the uniform size-matched sample defined above.
 - 11: Construct $Z_{b,\text{tr}} = [z_{b\ell}]_{\ell \in L_b}$ and $y_{b,\text{tr}} \leftarrow (y_b^t)_{L_b}$.
 - 12: $\Theta_b \leftarrow \text{FITPREDICTOR}(L, b, \Theta_b, Z_{b,\text{tr}}, y_{b,\text{tr}}, \lambda_b, \Delta_b, n_b, \vartheta)$; $\Delta_b \leftarrow 0$.
-

4 Initialization

Calibration draws random pairs (i_ℓ, j_ℓ) and sets

$$\bar{q}_{\text{cal}} = \frac{1}{10,000} \sum_{\ell=1}^{10,000} \{Q + f(x_{i_\ell}, x_{j_\ell})\},$$

$$r = f_r \bar{q}_{\text{cal}}, \quad \phi = c_s = \lambda_c \bar{q}_{\text{cal}}.$$

Local variables for Initialize

Symbol	Dimension/domain	Definition
λ_c	scalar	Cost rate used to set ϕ and c_s .
f_r	scalar	Reservation coefficient; sets $r = f_r \bar{q}_{\text{cal}}$ (default 0.60).
(i_ℓ, j_ℓ)	pair in $[N]^2$	Random calibration pair.
d_γ	integer	Number of active curve dimensions.
p_{rewire}	scalar	Watts-Strogatz rewiring probability.
ν_ℓ	integer	Frequency for active curve coordinate ℓ .
ψ_ℓ	scalar	Phase for active curve coordinate ℓ .
A_0	$\mathbb{R}^{d \times d}$	Raw matrix used to construct A .
Σ_x	$\mathbb{R}^{d \times d}$	Empirical agent type second moment.
B_0	$\mathbb{R}^{d \times d}$	Raw zero-trace symmetric boundary matrix.
B_{raw}	$\mathbb{R}^{d \times d}$	Orthogonalized pre-normalization boundary matrix.
$Z_{b,\text{seed}}, y_{b,\text{seed}}$	matrix, $\mathbb{R}^{n_b^0}$	Broker seed fitting batch using the selected input map.

Local variables for InitializeAgent

Symbol	Dimension/domain	Definition
σ_x	scalar	Agent type noise scale.
u_i	scalar	Agent i curve position.
ϵ_i	$\mathbb{R}^d$	Agent type noise for agent i .
π_{ij}^{ent}	scalar	Entrant attachment probability.
$\mathcal{V}_i^{\text{ent}}$	subset of $[N] \setminus \{i\}$	Entrant attachment set.

Algorithm 5 InitializeAgent

Require: State S , agent slot i , period counter t , parameters ϑ .

Ensure: Agent slot i is initialized in S .

- 1: Draw $u_i \sim \text{Unif}[0, 1]$ and $\epsilon_i \sim \mathcal{N}(0, \sigma_x^2 I_d/d)$.
 - 2: $x_i^t \leftarrow (\gamma(u_i) + \epsilon_i) / \|\gamma(u_i) + \epsilon_i\|_2$.
 - 3: Reset $\mathcal{H}_i^t = (\Xi_i^t, y_i^t, n_i^t)$ to zero observations, $\bar{q}_i^t$, C_i^t , χ_i^t , M_i^t , and Δ_i^t .
 - 4: Initialize the selected predictor to the constant Q : for NN, draw W_{1i}^t with He initialization and set $b_{1i}^t = 0$, $w_{2i}^t = 0$, $b_{2i}^t = Q$, $\mathcal{O}_i^t = 0$; for Ridge, set $\alpha_i^t = Q$ and $\beta_i^t = 0$.
 - 5: **if** $t = 0$ **then**
 - 6: $s_i^t \leftarrow (0, 0)$.
 - 7: **else**
 - 8: Remove i from Roster^t , D^t , and M_j^t for all j ; delete all incident edges in G^t ; set $\bar{q}_{ji}^t, C_{ji}^t \leftarrow 0$ for all $j \neq i$.
 - 9: Sample $\mathcal{V}_i^{\text{ent}} \subset [N] \setminus \{i\}$ without replacement, $|\mathcal{V}_i^{\text{ent}}| = \min(\lfloor k_G/2 \rfloor, N-1)$, with

$$\pi_{ij}^{\text{ent}} \propto \exp\{-\|x_i^t - x_j^t\|_2^2\}.$$
 - 10: Add edges (i, j) to G^t for all $j \in \mathcal{V}_i^{\text{ent}}$.
 - 11: $s_{i,\text{self}}^t \leftarrow \begin{cases} |\mathcal{V}_i^{\text{ent}}|^{-1} \sum_{j \in \mathcal{V}_i^{\text{ent}}} s_{j,\text{self}}^t, & \mathcal{V}_i^{\text{ent}} \neq \emptyset, \\ 0, & \mathcal{V}_i^{\text{ent}} = \emptyset. \end{cases}$
 - 12: $s_{i,\text{broker}}^t \leftarrow \text{rep}_b^t$.
 - 13: Set $\mathbf{p}_i^t \leftarrow (0)$, treating entrant initialization as period 0 of its tenure.
-

Algorithm 6 Initialize

Require: Parameters ϑ and seed.

Ensure: Initial state S^0 .

- 1: Draw curve frequencies $\nu_\ell \sim \text{Unif}\{1, \dots, 5\}$ and phases $\psi_\ell \sim \text{Unif}[0, 2\pi)$ for $\ell = 1, \dots, d_\gamma$, defining γ .
- 2: Allocate empty agent and broker containers in S^0 and store γ .
- 3: **for** $i = 1, \dots, N$ **do**
- 4: INITIALIZEAGENT($S^0, i, 0, \vartheta$).
- 5: Draw $u_c \sim \text{Unif}[0, 1]$ and $\epsilon_c \sim \mathcal{N}(0, \sigma_x^2 I_d/d)$; set $c \leftarrow [\gamma(u_c) + \epsilon_c] / \|\gamma(u_c) + \epsilon_c\|_2$.
- 6: Draw $A_0 \in \mathbb{R}^{d \times d}$ iid standard normal and set $A \leftarrow A_0^\top A_0 \cdot d / \text{tr}(A_0^\top A_0)$.
- 7: $\Sigma_x \leftarrow N^{-1} \sum_i x_i^0 (x_i^0)^\top$.
- 8: Draw symmetric zero-trace B_0 and set

$$B_{\text{raw}} \leftarrow B_0 - \frac{\text{tr}(\Sigma_x B_0 \Sigma_x A)}{\text{tr}(\Sigma_x A \Sigma_x A)} A, \quad B \leftarrow B_{\text{raw}} / \|B_{\text{raw}}\|_F.$$

```

9: Calibrate  $(\bar{q}_{\text{cal}}, r, \phi, c_s)$ .
10: Build  $G^0$  as a Watts-Strogatz graph on  $[N]$  with degree  $k_G$  and rewiring probability  $p_{\text{rewire}}$ ; add isolated broker node  $N + 1$ .
11: Initialize  $\mathcal{H}_b^0$  and the selected broker predictor to the constant  $Q$  using the corresponding NN or Ridge initialization above; set  $\text{rep}_b^0 = 0$ .
12: Sample  $\text{Roster}^0 \subset [N]$  uniformly with  $|\text{Roster}^0| = \lceil \alpha_R N \rceil$ ; add broker edges  $(i, N + 1)$  for all roster members.
13: for each unordered non-broker edge  $\{i, j\} \in \mathcal{E}_G^0$  do
14:   Draw  $q_{ij}$  once; append  $(x_j^0, q_{ij})$  to  $\mathcal{H}_i^0$  and  $(x_i^0, q_{ij})$  to  $\mathcal{H}_j^0$ ; update  $\bar{q}_{ij}^0, C_{ij}^0, \bar{q}_{ji}^0, C_{ji}^0$ .
15: Let  $\mathcal{B}_{\text{seed}} \leftarrow \{\{i, j\} \in \mathcal{E}_G^0 : i, j \in \text{Roster}^0\}$ ; shuffle and keep at most 100 edges.
16: for each  $\{i, j\} \in \mathcal{B}_{\text{seed}}$  do
17:   Append  $(x_i^0, x_j^0, q_{ij})$  to  $\mathcal{H}_b^0$ ; increment  $n_b^0$ .
18:   If  $\mathbf{L} = \text{Ridge}$  and  $a_b = \text{single-principal}$ , draw and store  $r_\ell \sim \text{Unif}\{1, 2\}$  for this observation.
19: if  $n_b^0 > 0$  then
20:   Set  $\text{rep}_b^0$  to mean broker seed output.
21: else
22:   Set  $\text{rep}_b^0 \leftarrow 0$ .
23: Set  $s_{i,\text{self}}^0$  to mean own seed output and  $s_{i,\text{broker}}^0 \leftarrow \text{rep}_b^0$ .
24: Set  $\mathbf{p}_i^0 \leftarrow (n_i^0)$  for every agent and  $\mathbf{p}_b^0 \leftarrow (n_b^0)$ , so initialization is period 0; set  $\Delta_i^0 \leftarrow n_i^0$  and  $\Delta_b^0 \leftarrow n_b^0$ .
25: for each  $i$  with  $n_i^0 > 0$  do
26:    $\Theta_i^0 \leftarrow \text{FITPREDICTOR}(\mathbf{L}, a, \Theta_i^0, \Xi_i^0, y_i^0, \lambda_a, n_i^0, n_i^0, \vartheta)$ ;  $\Delta_i^0 \leftarrow 0$ .
27: if  $n_b^0 > 0$  then
28:    $Z_{b,\text{seed}} \leftarrow [z_{b\ell}]_{\ell=1}^{n_b^0}$ ,  $y_{b,\text{seed}} \leftarrow y_b^0$ .
29:    $\Theta_b^0 \leftarrow \text{FITPREDICTOR}(\mathbf{L}, b, \Theta_b^0, Z_{b,\text{seed}}, y_{b,\text{seed}}, \lambda_b, n_b^0, n_b^0, \vartheta)$ ;  $\Delta_b^0 \leftarrow 0$ .
30: return  $S^0$ .

```

5 Channel Choice

Local variables for ChooseChannel

Symbol	Dimension/domain	Definition
U_{self}	scalar	Self-channel score.
U_{broker}	scalar	Broker-channel score.

Algorithm 7 ChooseChannel

Require: Agent i state (s_i, χ_i) , broker reputation rep_b , RNG.

```

1:  $U_{\text{self}} \leftarrow s_{i,\text{self}}$ .
2:  $U_{\text{broker}} \leftarrow \chi_i s_{i,\text{broker}} + (1 - \chi_i) \text{rep}_b$ .
3: if  $U_{\text{self}} > U_{\text{broker}}$  then
4:   return self.
5: if  $U_{\text{broker}} > U_{\text{self}}$  then
6:   return broker.
7: return self w.p.  $1/2$ , else broker  $\triangleright U_{\text{self}} = U_{\text{broker}}$ .

```

6 One Period

<i>Local variables for PeriodUpdate</i>		
Symbol	Dimension/domain	Definition
K	integer	Active-demand bound.
p_{demand}	scalar	Per-position demand probability.
α_R	scalar	Standing broker roster share.
p_{roster}	scalar	Roster-churn probability.
η_{exit}	scalar	Agent exit probability.
Δ_{net}	integer	Network-measure interval.

Algorithm 8 PeriodUpdate

Require: Previous state S^{t-1} , period t , parameters ϑ .

Ensure: Updated state S^t , accepted records $\mathcal{R}^t$, and **pre-turnover measure row** Y^t .

- 1: Clear accumulators, set $M_i^t \leftarrow \emptyset$ for all i , and set $D^t \leftarrow \emptyset$.
- 2: Refresh roster: remove each roster member with probability p_{roster} , then replenish uniformly to size $\lceil \alpha_R N \rceil$.
- 3: $\mathcal{E}_G^t \leftarrow (\mathcal{E}_G^t \setminus \{\{i, N+1\} : i \in [N]\}) \cup \{\{i, N+1\} : i \in \text{Roster}^t\}$.
- 4: $\mathcal{D}^t \leftarrow \emptyset$.
- 5: **for** $i = 1, \dots, N$ **do**
- 6: Draw $d_i^t \sim \text{Binomial}(K, p_{\text{demand}})$.
- 7: **if** $d_i^t > 0$ **then**
- 8: $z_i^t \leftarrow \text{CHOOSECHANNEL}(i, \text{rep}_b^{t-1})$.
- 9: Append (i, z_i^t, d_i^t) to $\mathcal{D}^t$.
- 10: **if** $z_i^t = \text{broker}$ **then**
- 11: Add i to D^t .
- 12: $\mathcal{E}_G^t \leftarrow (\mathcal{E}_G^t \setminus \{\{i, N+1\} : i \in [N]\}) \cup \{\{i, N+1\} : i \in \text{Roster}^t \cup D^t\}$.
- 13: $\text{TRAINPREDICTORS}(S, t, \vartheta)$.
- 14: $\mathcal{R}^t \leftarrow \text{OFFERMARKET}(S, \mathcal{D}^t, \vartheta)$.
- 15: Synchronize broker-node edges to Roster^t , D^t , and current broker-channel matches.
- 16: $\text{UPDATESATISFACTION}(S, \mathcal{D}^t, \mathcal{R}^t)$.
- 17: **if** $D^t \neq \emptyset$ **then**
- 18: $\text{rep}_b^t \leftarrow |D^t|^{-1} \sum_{i \in D^t} s_{i, \text{broker}}^t$.
- 19: **else**
- 20: $\text{rep}_b^t \leftarrow \text{rep}_b^{t-1}$.
- 21: **if** $t \bmod \Delta_{\text{net}} = 0$ **then**
- 22: **Recompute broker betweenness, Burt constraint, and effective size on the post-matching, pre-turnover graph** G^t .
- 23: $Y^t \leftarrow \text{COLLECTMEASURES}(S^t, \mathcal{R}^t, \vartheta)$ **on the post-matching, pre-turnover state.**
- 24: Append n_i^t to $\mathbf{p}_i^t$ for every agent and append n_b^t to $\mathbf{p}_b^t$, recording the end-of-period training-window boundaries.
- 25: **for** $i = 1, \dots, N$ **do**
- 26: **if** agent i exits with probability η_{exit} **then**
- 27: $\text{INITIALIZEAGENT}(S^t, i, t, \vartheta)$.
- 28: Synchronize broker-node edges to the post-turnover roster, clients, and active matches.
- 29: **return** $(S^t, \mathcal{R}^t, Y^t)$.

7 Offer Market

Local variables for OfferMarket

Symbol	Dimension/domain	Definition
$\hat{q}_i^t$	$\mathbb{R}^d \rightarrow \mathbb{R}$	Agent predictor.
$\hat{q}_b^t$	$\mathbb{R}^{d_b} \rightarrow \mathbb{R}$	Broker predictor.
$\mathcal{N}_G^t(i)$	subset of $[N]$	Non-broker graph neighbors of i .
U_i^t	subset of $[N]$	Self-search stranger pool independently sampled for demander i .
$n_{\text{strangers}}$	integer	Per-demander stranger-pool size.
$v_{i \rightarrow j}^{\text{self}, t}$	scalar	Self-offer score.
$\mathcal{U}_{b,i}$	subset of $[N]$	Candidate set for broker-client demander i .
$v_{b,i \rightarrow j}^t$	scalar	Score used to order brokered candidates for demander i .
Ω_i	integer	Residual offer quota for demander i .
Ω	$\mathbb{Z}_{\geq 0}^N$	Residual quota vector.
O^t	map	Directed offer book.
O_{ij}^t	offer record	Record (i, j, z, v) for offer $i \rightarrow j$.
$\mathcal{P}_O^t$	set	Unordered pairs represented in O^t .
$\mathcal{U}_i^{\text{self}}$	subset of $[N]$	Self-search candidate set.
$\mathcal{I}_i^{\text{self}}$	list	Self-search candidates sorted by score.
$\mathcal{I}_{b,i}$	list	Brokered candidates for client i , sorted by $v_{b,i \rightarrow j}^t$.
$\mathcal{B}_b^t$	list	Selected directed broker offers.
$u_{i \rightarrow j}^{\text{self}}, u_{b,i \rightarrow j}^{\text{local}}, u_{b,i \rightarrow j}^{\text{global}}$	$[0, 1]$	Independent random secondary keys used only within equal-score rankings.
o, o', o_{ij}, o_{ji}	offer records	Offer records; fields include $o.z$ and $o.from$.
z^*	channel	Accepted-record channel.
ι	$\{0, 1\}$	Return value of AddOffer.

The symmetric broker score is

$$v_b^t(i, j) = \text{PREDICTBROKER}(i, j).$$

Let $\mathcal{N}_G^t(i) = \{j \in [N] : (i, j) \in \mathcal{E}_G^t\}$ exclude the broker node. Agent i assesses any presented candidate as

$$a_{i \rightarrow j}^t = \begin{cases} \hat{q}_{ij}^t, & j \in \mathcal{N}_G^t(i), C_{ij}^t > 0, \\ \text{PREDICTAGENT}(i, x_j^t), & \text{otherwise.} \end{cases}$$

Self-search makes this score available only for neighbors and non-neighbors in the demander-specific stranger pool:

$$v_{i \rightarrow j}^{\text{self}, t} = \begin{cases} a_{i \rightarrow j}^t, & j \in \mathcal{N}_G^t(i) \cup (U_i^t \setminus \mathcal{N}_G^t(i)), \\ \perp, & \text{otherwise.} \end{cases}$$

Thus direct pair means replace predictor extrapolations when pair-specific experience exists; $\perp$ denotes no valid self-search score. The receiver-side acceptance score for a one-sided offer is

$$w_{j \leftarrow i}^t = \begin{cases} \hat{q}_{ji}^t, & (i, j) \in \mathcal{E}_G^t, C_{ji}^t > 0, \\ \text{PREDICTAGENT}(j, x_i^t), & \text{otherwise.} \end{cases}$$

The market creates at most d_i^t outgoing offers from demander i . Receivers have no per-period capacity constraint; acceptance is pairwise, so a receiver may accept multiple offers in the same period. A demanded position can remain unmatched if no acceptable offer is formed.

Algorithm 9 AddOffer

Require: Offer book O^t , unordered-pair set $\mathcal{P}_O^t$, offer (i, j, z, v) .

- 1: **if** O_{ij}^t is defined **then**
 - 2: **return** 0.
 - 3: $O_{ij}^t \leftarrow (i, j, z, v)$.
 - 4: **if** O_{ji}^t is undefined **then**
 - 5: $\mathcal{P}_O^t \leftarrow \mathcal{P}_O^t \cup \{\{i, j\}\}$.
 - 6: **return** 1.
-

Algorithm 10 BrokerCandidates

Require: Broker-client demander i , state S , broker service S_b .

Ensure: Candidate set $\mathcal{U}_{b,i}$ and score function $v_{b,i \rightarrow j}^t$.

- 1: **if** $S_b = \text{full}$ **then**
 - 2: $\mathcal{U}_{b,i} \leftarrow \mathcal{A}_b^t \setminus (\{i\} \cup M_i^t)$ and $v_{b,i \rightarrow j}^t \leftarrow v_b^t(i, j)$.
 - 3: **else if** $S_b = \text{assessment-only}$ **then**
 - 4: Draw $U_i^t \subset [N]$ uniformly without replacement, $|U_i^t| = \min(n_{\text{strangers}}, N)$.
 - 5: $\mathcal{U}_{b,i} \leftarrow [\mathcal{N}_G^t(i) \cup (U_i^t \setminus \mathcal{N}_G^t(i))] \setminus (\{i\} \cup M_i^t)$ and $v_{b,i \rightarrow j}^t \leftarrow v_b^t(i, j)$.
 - 6: **else**
 - 7: $\mathcal{U}_{b,i} \leftarrow \mathcal{A}_b^t \setminus (\{i\} \cup M_i^t)$ and $v_{b,i \rightarrow j}^t \leftarrow a_{i \rightarrow j}^t$. $\triangleright$ access-only
 - 8: **return** $(\mathcal{U}_{b,i}, v_{b,i \rightarrow j}^t)$.
-

Algorithm 11 OfferMarket

Require: State S , demand records $\mathcal{D}^t$, parameters ϑ .

Ensure: Accepted records $\mathcal{R}^t$.

- 1: For all $(i, z_i, d_i) \in \mathcal{D}^t$, set residual offer quota $\Omega_i \leftarrow d_i$.
- 2: $O^t \leftarrow \emptyset$, $\mathcal{P}_O^t \leftarrow \emptyset$, $\mathcal{R}^t \leftarrow \emptyset$.
- 3: **for** each $(i, \text{self}, d_i) \in \mathcal{D}^t$ with $\Omega_i > 0$ **do**
- 4: Draw $U_i^t \subset [N]$ uniformly without replacement, $|U_i^t| = \min(n_{\text{strangers}}, N)$.
- 5: $\mathcal{U}_i^{\text{self}} \leftarrow \mathcal{N}_G^t(i) \cup (U_i^t \setminus \mathcal{N}_G^t(i))$.
- 6: Remove from $\mathcal{U}_i^{\text{self}}$ the elements of $\{i\} \cup M_i^t$.
- 7: Draw independent $u_{i \rightarrow j}^{\text{self}} \sim \text{Unif}(0, 1)$ for each feasible j .
- 8: $\mathcal{I}_i^{\text{self}} \leftarrow \text{argsort}_{j \in \mathcal{U}_i^{\text{self}}: v_{i \rightarrow j}^{\text{self}, t} > r}(-v_{i \rightarrow j}^{\text{self}, t}, u_{i \rightarrow j}^{\text{self}})$.
- 9: **for** the first $\min(\Omega_i, |\mathcal{I}_i^{\text{self}}|)$ agents j in $\mathcal{I}_i^{\text{self}}$ **do**
- 10: $\text{ADDOFFER}(O^t, \mathcal{P}_O^t, i, j, \text{self}, v_{i \rightarrow j}^{\text{self}, t})$.
- 11: $\mathcal{A}_b^t \leftarrow \text{Roster}^t \cup D^t$.
- 12: $D_+^t \leftarrow \{i : (i, \text{broker}, d_i) \in \mathcal{D}^t, \Omega_i > 0\}$.
- 13: $\mathcal{B}_b^t \leftarrow \emptyset$.
- 14: **for** each $i \in D_+^t$ **do**
- 15: $(\mathcal{U}_{b,i}, v_{b,i \rightarrow j}^t) \leftarrow \text{BROKERCANDIDATES}(i, S, S_b)$.
- 16: Draw independent $u_{b,i \rightarrow j}^{\text{local}} \sim \text{Unif}(0, 1)$ for each feasible j .
- 17: $\mathcal{I}_{b,i} \leftarrow \text{argsort}_{j \in \mathcal{U}_{b,i}: v_{b,i \rightarrow j}^t > r}(-v_{b,i \rightarrow j}^t, u_{b,i \rightarrow j}^{\text{local}})$.

```

18:   for the first  $\min(\Omega_i, |\mathcal{I}_{b,i}|)$  agents  $j$  in  $\mathcal{I}_{b,i}$  do
19:     Append  $(i, j, v_{b,i \rightarrow j}^t)$  to  $\mathcal{B}_b^t$ .
20: Draw a new independent  $u_{b,i \rightarrow j}^{\text{global}} \sim \text{Unif}(0, 1)$  for each  $(i, j, v) \in \mathcal{B}_b^t$ .
21: Sort  $\mathcal{B}_b^t$  lexicographically by  $(-v, u_{b,i \rightarrow j}^{\text{global}})$ .
22: for each  $(i, j, v) \in \mathcal{B}_b^t$  do
23:   if  $\Omega_i > 0$  then
24:      $\iota \leftarrow \text{ADDOFFER}(O^t, \mathcal{P}_O^t, i, j, \text{broker}, v)$ .
25:      $\Omega_i \leftarrow \Omega_i - \iota$ .
26: for each  $\{i, j\} \in \mathcal{P}_O^t$  do
27:    $o_{ij} \leftarrow O_{ij}^t$  if defined, else  $\emptyset$ ;  $o_{ji} \leftarrow O_{ji}^t$  if defined, else  $\emptyset$ .
28:   if  $o_{ij} \neq \emptyset$  and  $o_{ji} \neq \emptyset$  then
29:      $\text{FINALIZEMATCH}(S, \mathcal{R}^t, o_{ij}, o_{ji})$ .
30:   else if  $o_{ij} \neq \emptyset$  and  $w_{j \leftarrow i}^t > r$  then
31:      $\text{FINALIZEMATCH}(S, \mathcal{R}^t, o_{ij}, \emptyset)$ .
32:   else if  $o_{ji} \neq \emptyset$  and  $w_{i \leftarrow j}^t > r$  then
33:      $\text{FINALIZEMATCH}(S, \mathcal{R}^t, o_{ji}, \emptyset)$ .
34: return  $\mathcal{R}^t$ .

```

Algorithm 12 FinalizeMatch

Require: State S , accepted-record list $\mathcal{R}^t$, offer $o = (i, j, z, v)$, optional reverse offer o' .

```

1: if  $j \in M_i^t$  then
2:   return .
3: Draw  $q_{ij}^t = Q + f(x_i^t, x_j^t) + \varepsilon$ ,  $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ .
4: Append columns  $x_j^t$  to  $\Xi_i$  and  $x_i^t$  to  $\Xi_j$ ; append  $q_{ij}^t$  to  $y_i$  and  $y_j$ ; increment  $n_i$  and  $n_j$ .
5: Update  $\bar{q}_{ij}, C_{ij}$  and  $\bar{q}_{ji}, C_{ji}$ .
6: Add edge  $(i, j)$  to  $G^t$ ; add  $j$  to  $M_i^t$  and  $i$  to  $M_j^t$ .
7: if  $z = \text{broker}$  or  $(o' \neq \emptyset$  and  $o'.z = \text{broker})$  then
8:   Append the two party types  $x_i^t$  and  $x_j^t$  to  $X_{b,1}$  and  $X_{b,2}$ ; append  $q_{ij}^t$  to  $y_b$ ; increment  $n_b$ .
9:   If  $\mathbf{L} = \text{Ridge}$  and  $a_b = \text{single-principal}$ , draw and store  $r_\ell \sim \text{Unif}\{1, 2\}$  for this observation.
10:   $z^* \leftarrow \text{broker}$ .
11: else
12:   $z^* \leftarrow \text{self}$ .
13: Append accepted record  $(i, j, z^*, q_{ij}^t, v, o, o')$  to  $\mathcal{R}^t$ .

```

8 Satisfaction and Measures

Local variables for satisfaction

Symbol	Dimension/domain	Definition
Y_i	scalar	Agent i period satisfaction numerator.
$n_{i,b}$	integer	Accepted broker offers credited to demander i .
$\bar{q}_i^t$	scalar	Channel-specific satisfaction input.
ζ	accepted record	Iterator over $\mathcal{R}^t$.
$\mathbf{1}\{\cdot\}$	indicator	Equals 1 when its condition is true and 0 otherwise.

Algorithm 13 UpdateSatisfaction

Require: State S , demand records $\mathcal{D}^t$, accepted records $\mathcal{R}^t$.

```

1: for each  $(i, z_i, d_i) \in \mathcal{D}^t$  do
2:    $Y_i \leftarrow \sum_{\zeta \in \mathcal{R}^t} \sum_{o \in \{\zeta.o_1, \zeta.o_2\}} \mathbf{1}\{o \neq \emptyset, o.\text{from} = i\} \zeta.q.$ 
3:    $n_{i,b} \leftarrow \sum_{\zeta \in \mathcal{R}^t} \sum_{o \in \{\zeta.o_1, \zeta.o_2\}} \mathbf{1}\{o \neq \emptyset, o.\text{from} = i, o.z = \text{broker}\}.$ 
4:   if  $z_i = \text{self}$  then
5:      $\tilde{q}_i^t \leftarrow Y_i/d_i - c_s.$ 
6:   else
7:      $\tilde{q}_i^t \leftarrow (Y_i - \phi n_{i,b})/d_i; \chi_i^t \leftarrow 1.$ 
8:    $s_{i,z_i}^t \leftarrow (1 - \omega)s_{i,z_i}^{t-1} + \omega \tilde{q}_i^t.$ 

```

CollectMeasures. COLLECTMEASURES($S^t, \mathcal{R}^t, \vartheta$) records the measures defined in Appendix B, Section 8, without changing the simulation state.

Appendix B

Model Specifications for an ABM of Brokerage in Matching Markets

This document specifies an agent-based model of brokered matching.

The model is a discrete-time simulation of a matching market with *agents* and a *broker*. Agents seeking pairwise matches either search on their own or outsource to the broker. All economic quantities (match output, fees, surplus) are in the same payoff units. All agents use heuristic decision rules. No agent holds beliefs about other agents' strategies.

The model specification covers agents (§1), the matching problem (§2), how agents learn to predict expected match output (§3), match economics (§4), network structure and agent turnover (§5), search and broker access (§6), the outsourcing decision (§7), and period measures (§8). Baseline parameter values and reproducibility are documented in Part II (§9).

Part I. Model Specification

1. Agents and Time

The model runs for T discrete periods (default 500). It has N agents (default 1000), a single broker, and an undirected network G over the agents and broker.

Each agent i is characterized by:

- **Type** $\mathbf{x}_i \in \mathbb{R}^d$: a fixed vector of observable characteristics, drawn as specified below and used by the matching function (§2).
- **Current-period matches** M_i^t : active bilateral relationships involving i in period t . Each counterparty can appear at most once in M_i^t ; incoming accepted offers are not capped.
- **Active demand bound** K : the maximum number of active demands an agent can initiate in a period (default $K = 5$; §6).
- **Experience history** $\mathcal{H}_i^t = \{(\mathbf{x}_j, q_{ij})\}$: (other party's type, realized match output) pairs from all matches involving i (§3a).
- **Satisfaction indices** $s_{i,c}^t$: one scalar per search channel $c \in \{\text{self}, \text{broker}\}$, used in the outsourcing decision (§7).

Agent Types

Type vectors are multidimensional but concentrated near a smooth one-dimensional curve on the unit sphere in $\mathbb{R}^d$. Let $u \in [0, 1]$ index position along the curve:

$$\gamma(u) = \frac{\tilde{\gamma}(u)}{\|\tilde{\gamma}(u)\|}, \quad \tilde{\gamma}_\ell(u) = \begin{cases} \sin(2\pi\nu_\ell u + \psi_\ell) & \ell = 1, \dots, d_\gamma \\ 0 & \ell = d_\gamma + 1, \dots, d. \end{cases}$$

The frequencies $\nu_\ell \sim \text{Unif}\{1, 2, 3, 4, 5\}$ and phases $\psi_\ell \sim \text{Unif}[0, 2\pi)$ are drawn once per simulation. The parameter $d_\gamma \leq d$ is the number of coordinates in which the curve varies.

Each agent is drawn at a random position $u_i \sim \text{Unif}[0, 1]$ on the curve, then perturbed:

$$\mathbf{x}_i = \frac{\gamma(u_i) + \boldsymbol{\epsilon}_i}{\|\gamma(u_i) + \boldsymbol{\epsilon}_i\|}, \quad \boldsymbol{\epsilon}_i \sim \mathcal{N}\left(\mathbf{0}, \frac{\sigma_x^2}{d} \mathbf{I}_d\right).$$

Noise is applied in all d dimensions, then the vector is re-normalized. The scale $\sigma_x/\sqrt{d}$ keeps the expected displacement from the curve approximately σ_x for any d .

Thus the curve remains one-dimensional for every d_γ . Increasing d_γ changes how many coordinates contain systematic curve variation; the remaining coordinates contain only perturbation noise.

Broker

The broker is a permanent node in G and has no type vector. Its state is:

- **Experience history** $\mathcal{H}_b^t = \{(\mathbf{x}_i, \mathbf{x}_j, q_{ij})\}$: (party 1 type, party 2 type, realized match output) triples from mediated matches (§3c). The party labels have no directional meaning because the broker's pair features are symmetric.
- **Roster** Roster^t : the broker's standing access base (§6b).
- **Current clients** D^t : agents who outsource to the broker in period t (§6b).
- **Reputation** rep^t : the average broker satisfaction of current clients (§7).

2. The Matching Problem

Each potential pair's expected match output combines two sources: the agents' average general quality and a match-specific complementarity between their types. The parameter ρ moves the matching problem from pure complementarity to pure general quality. The parameter δ makes complementarity harder to predict by increasing the difference between otherwise similar pairs on opposite sides of a boundary in type-pair space. Expected and realized match output are symmetric in the two agents.

2a. Match Output

Let q_{ij} be the realized per-period output of a match between agents i and j , measured in payoff units:

$$q_{ij} = Q + f(\mathbf{x}_i, \mathbf{x}_j) + \varepsilon_{ij}, \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_\varepsilon^2).$$

The conditional expected output is $E[q_{ij} \mid \mathbf{x}_i, \mathbf{x}_j] = Q + f(\mathbf{x}_i, \mathbf{x}_j)$. The systematic component f is fully determined by the two types and by matching-function objects fixed at initialization. The

independent, mean-zero shock ε_{ij} creates match-specific variation that neither the agents nor the broker can predict.

The function $f : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ is unknown to all agents:

$$f(\mathbf{x}_i, \mathbf{x}_j) = \frac{1}{s_{\text{mix}}(\rho, \delta)} \left[\sqrt{\rho} \tilde{U}_{ij} + \sqrt{1 - \rho} \tilde{W}_{ij}(\delta) \right].$$

The tildes indicate that general quality (§2b) and match-specific complementarity (§2c) are centered and scaled using fixed moments. These moments are estimated once from independently generated matching environments. Within each environment, they are calculated across all unordered pairs of distinct agents; they are then averaged across environments and held fixed in every simulation. Individual markets are not rescaled, so they retain ordinary between-market variation.

Let $\kappa(\delta)$ denote the fixed covariance between general quality and complementarity after each component is standardized. The common adjustment

$$s_{\text{mix}}^2(\rho, \delta) = 1 + 2\sqrt{\rho(1 - \rho)} \kappa(\delta)$$

accounts for this small covariance. Averaged across the independently generated environments, each standardized component and the resulting systematic component f have within-market variance one for every ρ and δ . Because Q is constant, expected match output has the same variance. This defines the payoff unit. In these units, $Q = 2.84$ and $\sigma_\varepsilon = 0.28$.

Because the two components have the same marginal variance on this scale, $\rho \in [0, 1]$ is general quality's share of their summed marginal variances, while $1 - \rho$ is complementarity's share. Thus $\rho = 0$ gives pure complementarity and $\rho = 1$ gives pure general quality. At $\rho = 1$, δ has no effect.

2b. Agent General Quality

$\mathbf{c}$ is an ideal type vector. An agent's general-quality score, $\mathbf{x}_i^\top \mathbf{c}$, measures its alignment with that ideal. Pair-level general quality is the average of the two agents' scores:

$$U_{ij} = \frac{1}{2}(\mathbf{x}_i^\top \mathbf{c} + \mathbf{x}_j^\top \mathbf{c}), \quad \tilde{U}_{ij} = \frac{U_{ij} - \mu_U}{\sigma_U}.$$

The ideal type $\mathbf{c} \in \mathbb{R}^d$ is drawn at initialization as a perturbation of a random point on the agent type curve:

$$u_c \sim \text{Unif}[0, 1], \quad \boldsymbol{\epsilon}_c \sim \mathcal{N}\left(\mathbf{0}, \frac{\sigma_x^2}{d} \mathbf{I}_d\right), \quad \mathbf{c} = \frac{\boldsymbol{\gamma}(u_c) + \boldsymbol{\epsilon}_c}{\|\boldsymbol{\gamma}(u_c) + \boldsymbol{\epsilon}_c\|_2}.$$

The normalization places $\mathbf{c}$ on the same unit sphere as agent types. The perturbation therefore changes the ideal direction without changing its magnitude.

2c. Match-Specific Complementarity

Define base complementarity, signed complementarity, and difficulty-adjusted complementarity as

$$V_{ij} = \mathbf{x}_i^\top \mathbf{A} \mathbf{x}_j, \quad H_{ij} = \text{sign}(\mathbf{x}_i^\top \mathbf{B} \mathbf{x}_j) V_{ij}, \quad W_{ij}(\delta) = V_{ij} + \delta H_{ij}.$$

The normalized complementarity used in the matching function is

$$\widetilde{W}_{ij}(\delta) = \frac{W_{ij}(\delta) - \mu_W(\delta)}{\sigma_W(\delta)}.$$

$\mathbf{A}$ is a symmetric positive definite matrix that gives a smooth bilinear complementarity geometry. Because V_{ij} depends jointly on both types, it cannot in general be represented as the sum of separate agent scores. A second symmetric matrix, $\mathbf{B}$, assigns pairs to two regions of type-pair space. Both matrices are fixed for the duration of the simulation.

Base complementarity. Draw $\mathbf{M}_A \in \mathbb{R}^{d \times d}$ with iid $\mathcal{N}(0, 1)$ entries and set

$$\mathbf{A} = \mathbf{M}_A^\top \mathbf{M}_A \cdot \frac{d}{\text{tr}(\mathbf{M}_A^\top \mathbf{M}_A)}.$$

Thus $\mathbf{A}$ is symmetric positive definite and $\text{tr}(\mathbf{A}) = d$.

The trace normalization prevents arbitrary scale differences in $\mathbf{A}$ across realizations. The fixed pair-level normalization described in §2a then places complementarity on the common scale used in the matching function.

Difficulty parameter. Equivalently, write $W_{ij}(\delta) = g(\mathbf{x}_i, \mathbf{x}_j)V_{ij}$, where a second symmetric matrix $\mathbf{B} \in \mathbb{R}^{d \times d}$ defines

$$g(\mathbf{x}_i, \mathbf{x}_j) = 1 + \delta \cdot \text{sign}(\mathbf{x}_i^\top \mathbf{B} \mathbf{x}_j)$$

where $\delta \in [0, 1]$ (default 0.5).

The multiplier is $1 + \delta$ when $\mathbf{x}_i^\top \mathbf{B} \mathbf{x}_j > 0$, $1 - \delta$ when it is negative, and 1 when it is zero. At $\delta = 0$, the boundary between the two regions has no effect. For $\delta > 0$, pairs on opposite sides of the boundary can have sharply different expected output even when their types are otherwise similar. Increasing δ increases this difference, making complementarity harder to learn. Normalizing $W_{ij}(\delta)$ separately for each δ removes the accompanying change in dispersion. Thus δ changes the shape and difficulty of complementarity, not its variance.

Boundary construction. Matrix $\mathbf{B}$ is made orthogonal to $\mathbf{A}$ over the realized type distribution. This ensures that δ introduces a distinct boundary rather than merely amplifying the base complementarity pattern. Let

$$\Sigma_x = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_i \mathbf{x}_i^\top$$

be the uncentered second-moment matrix of the realized type vectors.

Draw $\mathbf{Z} \in \mathbb{R}^{d \times d}$ with iid $\mathcal{N}(0, 1)$ entries and form a symmetric zero-trace matrix:

$$\tilde{\mathbf{B}} = \frac{\mathbf{Z} + \mathbf{Z}^\top}{2}, \quad \mathbf{B}_0 = \tilde{\mathbf{B}} - \frac{\text{tr}(\tilde{\mathbf{B}})}{d} \mathbf{I}_d.$$

Define the weighted inner product

$$\langle \mathbf{M}, \mathbf{N} \rangle_{\Sigma_x} = \text{tr}(\Sigma_x \mathbf{M} \Sigma_x \mathbf{N}),$$

then remove the projection of $\mathbf{B}_0$ onto $\mathbf{A}$ and normalize:

$$\mathbf{B}_{\text{raw}} = \mathbf{B}_0 - \frac{\text{tr}(\Sigma_x \mathbf{B}_0 \Sigma_x \mathbf{A})}{\text{tr}(\Sigma_x \mathbf{A} \Sigma_x \mathbf{A})} \mathbf{A}, \quad \mathbf{B} = \frac{\mathbf{B}_{\text{raw}}}{\|\mathbf{B}_{\text{raw}}\|_F}.$$

This gives $\text{tr}(\Sigma_x \mathbf{B} \Sigma_x \mathbf{A}) = 0$ and $\|\mathbf{B}\|_F = 1$. Weighting by Σ_x removes overlap between the two bilinear scores where agents actually occur, rather than in unused regions of $\mathbb{R}^d$.

Symmetry of $\mathbf{A}$ and $\mathbf{B}$ makes $f(\mathbf{x}_i, \mathbf{x}_j) = f(\mathbf{x}_j, \mathbf{x}_i)$.

3. Learning

Before searching for matches, agents and the broker may update their prediction models on their accumulated histories. Updated models are used to score candidates (§6). The broker is eligible to refit every period. Agent i is eligible when $i \bmod 2 = t \bmod 2$, so each agent is eligible every other period. An eligible model refits only when its history is nonempty and it has received new observations since its previous fitting call. A parameter switch selects the prediction model. Neural-network learning is the default. Ridge regression is the alternate experiment.

3a. Prediction Models and Fitting

In the default model, both agents and the broker use a fully-connected network with one hidden layer, ReLU activations, and a single linear output:

$$\hat{q}(\mathbf{z}) = \mathbf{w}_2^\top \text{ReLU}(\mathbf{W}_1 \mathbf{z} + \mathbf{b}_1) + b_2$$

where $\mathbf{z}$ is the input feature vector and the remaining symbols are network weights and biases. Agents receive raw partner type vectors, the broker receives pairs of them.

Training rule. Network weights minimize MSE with the Adam optimizer on the current training window. Agents and the broker may have separate learning rates, $\eta_{\text{lr},a}$ and $\eta_{\text{lr},b}$. No explicit weight regularization is applied. At $t = 0$, each network sets $\mathbf{W}_1$ using He initialization, $\mathbf{b}_1 = \mathbf{0}$, $\mathbf{w}_2 = \mathbf{0}$, and $b_2 = Q$ (§2a). An untrained predictor therefore returns exactly Q for every input. Networks with seed observations then take $E_{\text{init},c}$ gradient steps on their seed histories, where $c \in \{a, b\}$.

Alternate Ridge rule. In the Ridge experiment, agents and the broker instead use

$$\hat{q}_i(\mathbf{x}_j) = \alpha_i + \beta_i^\top \mathbf{x}_j, \quad \hat{q}_b(\mathbf{x}_i, \mathbf{x}_j) = \alpha_b + \gamma_b^\top \psi(\mathbf{x}_i, \mathbf{x}_j).$$

Agent and broker fits minimize mean squared error plus λ_a and λ_b , respectively, times the squared Euclidean norm of the slope coefficients. Intercepts are not penalized. Raw type and pair features are used without sample standardization. Before its first fit, a Ridge predictor has zero slopes and intercept Q . The reporting simulations use $\lambda_a = 0.003$ and $\lambda_b = 0.001$.

Subsequent NN updates warm-start from the previous weights and persistent Adam moment state. The number of gradient steps adapts to the ratio of new observations to total history:

$$E_{c,t} = \max\left(E_{\text{steps},c}, \left\lceil E_{\text{init},c} \cdot \frac{n_{\text{new}}}{n_{\text{total}}} \right\rceil\right), \quad c \in \{a, b\}.$$

where n_{new} is the number of observations added since that model’s previous fitting call, $n_{\text{total}} = |\mathcal{H}^t|$ is full history size, and $E_{\text{steps},c}$ is the learner-specific minimum per-call step count. For an agent, n_{new} can span multiple periods because agents fit on an alternating schedule. Agents and the broker use the same functional rule but may have different parameter values. Ridge is solved directly at each fitting call. The reporting NN values are $\eta_{\text{lr},a} = 0.003$, $\eta_{\text{lr},b} = 0.03$, $E_{\text{init},a} = E_{\text{init},b} = 100$, and $E_{\text{steps},a} = E_{\text{steps},b} = 50$.

Initialization is period 0 for each learner. Its seed-history count is stored as the first period boundary. Each later boundary stores cumulative history at the end of a simulation period. A fitting call uses observations from the most recent $W_p = 40$ completed simulation periods. Before 40 simulation periods are available, the window also includes period 0. Once 40 are available, the period-0 seed history is excluded. If the selected history block exceeds $M_{\text{train}} = 2000$ observations, fitting deterministically keeps 2000 spaced observations across that block, including its oldest and newest observations. If the cap is one, it keeps the newest observation. The cap changes only the fitting sample, not the stored history or the period window. NN and Ridge use the same window, cap, fitting cadence, and histories.

3b. Agent i ’s Model

History. $\mathcal{H}_i^t = \{(\mathbf{x}_j, q_{ij})\}_{m=1}^{n_i}$ records the other party’s type and realized match output from every match involving i .

Input. The agent’s network takes the partner’s type as input: $\mathbf{z} = \mathbf{x}_j$.

The hidden width is $h_A = 2d$, so the baseline uses 16 hidden units.

3c. Broker’s Model

History. $\mathcal{H}_b^t = \{(\mathbf{x}_i, \mathbf{x}_j, q_{ij})\}_{m=1}^{n_b}$ records both parties’ types and realized match output from broker-mediated matches.

Input. The broker’s network takes a symmetric pair feature vector:

$$\psi(\mathbf{x}_i, \mathbf{x}_j) = \left[\mathbf{x}_i + \mathbf{x}_j; \text{vech}\left(\frac{\mathbf{x}_i \mathbf{x}_j^\top + \mathbf{x}_j \mathbf{x}_i^\top}{2}\right) \right].$$

The half-vectorization vech keeps the lower-triangular entries of the symmetric matrix, so the broker input dimension is $d_B = d + d(d+1)/2$, equal to 44 at $d = 8$.

The broker hidden width is $h_B = 8d$, so the baseline uses 64 hidden units.

Because $\psi(\mathbf{x}_i, \mathbf{x}_j) = \psi(\mathbf{x}_j, \mathbf{x}_i)$, each observed unordered pair contributes one training row.

The broker learns only from mediated matches. It does not observe self-search outcomes.

3d. Ridge Broker Ablations

Three separate Ridge-only ablations change the broker while leaving every agent’s local Ridge model unchanged. **Size-matched pooled history** uniformly samples the broker’s windowed and capped fitting data without replacement, using the smaller of its available sample size and the nearest-integer median sample size among agents with nonempty histories, with half-integers rounded up. **Single-principal history** randomly retains one endpoint when each broker observation is recorded,

fits $g_b(\mathbf{x}) = \alpha_b + \gamma_b^\top \mathbf{x}$, and scores a pair as $g_b(\mathbf{x}_i) + g_b(\mathbf{x}_j) - \bar{q}_b$, where $\bar{q}_b$ is the current fitting-sample output mean and defaults to Q before fitting. **Additive Ridge** uses $\mathbf{x}_i + \mathbf{x}_j$ without bilinear features. All use the calibrated agent and broker penalties, period window, observation cap, and fitting cadence.

4. Match Economics

Matches use a common reservation threshold and channel-specific cost accounting.

At initialization, 10,000 random agent pairs define the calibration mean $\bar{q}_{\text{cal}} = E[q]$. This calibration quantity scales r , ϕ , and c_s .

4a. Outside Options

All agents share a reservation threshold r , calibrated at initialization:

$$r = f_r \cdot \bar{q}_{\text{cal}}$$

where $\bar{q}_{\text{cal}}$ is the calibration mean and f_r is the reservation coefficient (default 0.60). Since ϕ and c_s do not depend on r (§4c), f_r may take any nonnegative value, including $f_r \geq 1$, and is swept as an analysis axis.

4b. Participation Constraints

Demanders emit offers only above the reservation threshold. Self-search uses the demander evaluation $\hat{q}_i(\mathbf{x}_j) > r$. Brokered search uses the broker pair prediction $\hat{q}_b(\{i, j\}) > r$. A one-sided offer is accepted only if the receiver’s evaluation exceeds r ; reciprocal offers are accepted automatically (§6d).

4c. Search Costs

The self-search cost and broker fee share one calibrated level:

$$\phi = c_s = \lambda_c \cdot \bar{q}_{\text{cal}}.$$

λ_c is the shared cost rate (default 0.05). The two symbols differ in payoff accounting:

- **Self-search** costs c_s on each requested self-search position, whether or not it is filled.
- **Brokerage** charges ϕ only on successful brokered placements.

5. Network Structure and Turnover

Agents interact through a single undirected network G that determines search opportunities and structural position.

5a. Network Initialization

G is initialized as a Watts-Strogatz small-world graph on the N agents (Watts & Strogatz, 1998), with ring degree $k_G = 6$, rewiring probability $p_{\text{rewire}} = 0.1$, and random agent order on the ring.

The broker is added as a permanent node with no type vector. Broker-node edges are synchronized to standing roster members, current-period broker clients, and agents currently engaged in broker-channel matches (§6b).

5b. Match Tie Formation

Each realized match adds an undirected i - j edge if one does not already exist. Ties persist until either node exits.

5c. Agent Turnover

Agents exit independently each period with probability η_{exit} (default 0.02). An exiting agent is removed everywhere. Its node slot is reused by an entrant with a fresh type drawn from the same type process (§1). The entrant receives $\lfloor k_G/2 \rfloor$ edges to existing agents sampled with probability proportional to $\exp(-\|\mathbf{x}_{i'} - \mathbf{x}_j\|^2)$.

6. Search

At the start of each period, each agent draws active relationship demand

$$d_i \sim \text{Binomial}(K, p_{\text{demand}}),$$

where K is the maximum number of active demands the agent can initiate in the period and p_{demand} defaults to 0.50. If $d_i > 0$, the agent chooses **one channel for the batch** (§7): self-search or broker.

6a. Self-Search

Agent i 's self-search candidate pool has two components:

Neighbors. For a direct network neighbor j with prior pair-specific experience, agent i uses $\bar{q}_{ij} = \frac{1}{n_{ij}} \sum q_{ij}^{(m)}$, where n_{ij} is the number of times i and j have matched. If the graph edge exists but $n_{ij} = 0$, the agent instead uses $\hat{q}_i(\mathbf{x}_j)$. Initial histories seed one observation for every initial graph edge. Unobserved neighbor edges arise after turnover because entrants attach structurally before matching.

Strangers. Each self-searching demander i independently samples a pool U_i^t of $\min(n_{\text{strangers}}, N)$ agents uniformly without replacement, where $n_{\text{strangers}} = 10$ (default). The demander evaluates pool members that are not current neighbors using $\hat{q}_i(\mathbf{x}_j)$ (§3b).

Agent i orders feasible self-search candidates by these evaluations and emits up to d_i directed offers to candidates above r . Equal evaluations are ordered uniformly at random, so an agent whose predictions are constant produces a random ranking rather than an undefined ordering.

6b. Broker Roster and Access

The broker maintains a **roster** of agents it can propose as counterparties when mediating matches. In each period, it also observes the current-client set D^t of agents who outsourced. Broker access for the period is:

$$\mathcal{A}_b^t = \text{Roster}^t \cup D^t.$$

Current clients expand broker access for the period but do not become standing roster members.

Initialization. The roster is seeded with a target size

$$R^* = \lceil \alpha_R N \rceil,$$

where the broker roster share α_R defaults to 0.20. The model draws R^* agents uniformly at random (default 200 at $N = 1000$). The broker’s history is seeded from existing roster-roster ties in the initial graph, capped at 100 observed ties.

Roster refresh. At the start of each period, each roster member independently exits with probability p_{roster} (default 0.02). If $\widetilde{\text{Roster}}^t$ is the post-churn roster,

$$\widetilde{\text{Roster}}^t = \{i \in \text{Roster}^{t-1} : u_i^t > p_{\text{roster}}\}, \quad u_i^t \stackrel{iid}{\sim} \text{Unif}[0, 1],$$

then the broker samples without replacement from $\{1, \dots, N\} \setminus \widetilde{\text{Roster}}^t$ until $|\text{Roster}^t| = R^*$, or until the population is exhausted. Broker-node edges are synchronized to the standing roster, current clients, and current broker-mediated matches (§5a).

6c. Broker-Mediated Search

For every unordered pair $\{i, j\}$ such that one side is a broker-client demander and the other side is in $\mathcal{A}_b^t$, the broker computes predicted match output. For each broker-client demander i , it ranks accessible counterparties above r , breaking equal predictions uniformly at random, and keeps the highest-ranked candidates up to i ’s residual active broker demand. Directed broker offers are then processed in a global score ranking with a new, independent random ordering within score ties. If both sides are broker-client demanders with remaining active demand, the same unordered pair can generate reciprocal broker offers.

The broker-service experiment separates access from assessment. The default **full-service broker** uses the broker access set $\mathcal{A}_b^t$ and the broker’s pair predictions. The **assessment-only broker** applies the broker’s pair predictions to the same candidate set an agent could reach through self-search: its network neighbors and an independently drawn stranger pool. The **access-only broker** presents candidates from $\mathcal{A}_b^t$, but the client orders them using its own assessments: its pair-specific mean for an experienced neighbor and its prediction model otherwise. In every mode, candidates below the score source’s reservation threshold are excluded.

These restrictions change only how brokered candidates are obtained and ordered. The broker continues to observe mediated outcomes and refit its model in all modes, although its predictions do not determine access-only offers. Offer acceptance, match formation, fees, satisfaction, reputation, learning histories, and network updating are unchanged. “Assessment-only” means that the broker adds no candidate access beyond ordinary self-search. It does not imply that every resulting match

joins existing neighbors, because ordinary self-search includes strangers. Likewise, access-only service can select a current neighbor when that neighbor belongs to the broker access set.

6d. Offer Acceptance and Match Formation

All self-search and broker offers enter one shared market, grouped by unordered pair.

1. If pair $\{i, j\}$ contains reciprocal offers $i \rightarrow j$ and $j \rightarrow i$, the relationship is accepted automatically.
2. If pair $\{i, j\}$ contains one directed offer $i \rightarrow j$, receiver j evaluates i using $\bar{q}_{ji}$ for known partners and $\hat{q}_j(\mathbf{x}_i)$ otherwise. The offer is accepted iff this value exceeds r .
3. Each accepted unordered pair forms at most one current-period relationship.

At market finalization, for each accepted match: 1. Realized output is drawn: $q_{ij} = Q + f(\mathbf{x}_i, \mathbf{x}_j) + \varepsilon_{ij}$. 2. Both parties add the observation to their histories: i adds $(\mathbf{x}_j, q_{ij})$ to $\mathcal{H}_i$; j adds $(\mathbf{x}_i, q_{ij})$ to $\mathcal{H}_j$. 3. If either directed offer used the broker channel, the broker adds $(\mathbf{x}_i, \mathbf{x}_j, q_{ij})$ to $\mathcal{H}_b$ once. 4. An edge is added between i and j in G (if not already present).

A demander can have at most d_i outgoing offers, but can receive any number of acceptable incoming offers. Thus K is an active-demand bound, not a capacity constraint on receiving offers.

Before the next period begins, match lists are cleared. The next period begins with no active relationships.

7. The Outsourcing Decision

7a. Satisfaction Tracking

Each agent i maintains a satisfaction index $s_{i,c}^t$ for each search channel $c \in \{\text{self}, \text{broker}\}$. For the chosen channel, the index updates once per active-demand period as an EWMA of realized match output net of search costs:

$$s_{i,c}^{t+1} = (1 - \omega) s_{i,c}^t + \omega \cdot \tilde{q}$$

where $\omega = 0.2$ and $\tilde{q}$ is the period input in the table. The denominator is requested demand d_i , so unfilled requests contribute zero output. A reciprocal relationship can update two channel-specific states, one for each directed offer.

Channel	Satisfaction input $\tilde{q}$
Self-search	$\frac{\sum q_{ij}}{d_i} - c_s$, summing over accepted directed offers made by i through self-search
Brokered	$\frac{\sum q_{ij} - \phi \cdot n_{i,\text{broker success}}}{d_i}$, summing over accepted directed offers made by i through the broker

Brokered failure gives $\tilde{q} = 0$, self-search failure gives $\tilde{q} = -c_s$.

$s_{i,\text{self}}^t$ covers all self-search, including known neighbors.

At initialization, self-satisfaction is the mean of seed match outcomes and broker-satisfaction is the broker’s seed-data reputation (§7c). Fresh entrants set self-satisfaction to mean neighbor self-satisfaction and broker-satisfaction to current broker reputation.

The first broker choice, regardless of placement success, switches the agent from broker reputation to its own $s_{i,\text{broker}}^t$ (§7b).

7b. Decision Rule

Each agent with $d_i > 0$ compares:

- $\text{score}_{\text{self}} = s_{i,\text{self}}^t$
- $\text{score}_{\text{broker}} = s_{i,\text{broker}}^t$ if the agent has tried the broker, otherwise rep_b^t .

The agent chooses the higher-score channel. Ties are resolved by fair coin flip.

7c. Broker Reputation

$$\text{rep}_b^{t+1} = \begin{cases} \frac{1}{|D^t|} \sum_{i \in D^t} s_{i,\text{broker}}^{t+1} & \text{if } D^t \neq \emptyset \\ \text{rep}_b^t & \text{otherwise} \end{cases}$$

where D^t is the set of agents who outsourced to the broker this period. With current clients, reputation is their mean post-update broker satisfaction; with no clients, it is unchanged. Reputation is initialized from the mean broker seed outcome.

8. Period Measures

Period measures are evaluated after current-period matching, satisfaction, and reputation updates, but before agent entry/exit turnover. Agent decisions do not depend on these measures.

Network Measures

Broker network-position measures use G , including the broker node (§5a). They are computed when $t \bmod \Delta_{\text{net}} = 0$ (default 20). Other period measures are recorded each period.

Betweenness centrality. Freeman betweenness (Freeman, 1977) for the broker node:

$$\text{BC}_b = \frac{1}{(n-1)(n-2)} \sum_{u \neq b} \sum_{v \neq u} \frac{\sigma_{uv}(b)}{\sigma_{uv}}$$

where $n = N + 1$, σ_{uv} is the number of shortest paths from u to v , and $\sigma_{uv}(b)$ is the number passing through b . The double sum counts undirected pairs from both directions, so the denominator is $(n-1)(n-2)$ (Brandes, 2001, p. 9).

Burt’s constraint. Broker ego-network constraint (Burt, 1992):

$$\text{Constraint}_b = \sum_j \left(p_{bj} + \sum_{h \neq b, j} p_{bh} p_{hj} \right)^2$$

where $p_{bj} = 1/d_b$ and $p_{hj} = 1/d_h$ (Everett & Borgatti, 2020). The indirect term uses intermediary h 's degree.

Effective size. Non-redundant broker contacts using the Borgatti (1997) binary-undirected simplification:

$$ES_b = d_b - \frac{2t_b}{d_b},$$

where d_b is the broker's degree and t_b is the number of ties among the broker's neighbors, excluding ties to the broker.

Prediction Quality

Prediction quality is evaluated on random candidate pairs and accepted directed offers:

Sample	Construction	Reported measures
Random candidate pairs	Up to 100 agents with non-empty histories; for each sampled agent i , $\min(40, N - 1)$ non-self partners j . Agent and broker predictions are evaluated against noiseless $Q + f(\mathbf{x}_i, \mathbf{x}_j)$. Both models use the same sample, which is deterministic conditional on seed and period.	Per-agent R^2 , RMSE, bias, and Spearman rank correlation, averaged across sampled agents.
Selected offers	Accepted directed offers by channel, self-search or brokered. A reciprocal relationship can represent two directed decisions.	Channel-specific R^2 , RMSE, bias, and Spearman rank correlation.

The definitions are

$$R^2 = 1 - \frac{\text{MSE}}{\text{Var}_{\text{pop}}(q)}, \quad \text{RMSE} = \sqrt{\frac{1}{n} \sum (\hat{q} - q)^2}, \quad \text{Bias} = \frac{1}{n} \sum (\hat{q} - q).$$

Rank correlation evaluates the strict ranking produced from predicted output. Candidates are ordered by predicted output, with each set of equal predictions ordered uniformly at random. Agent and broker tie-breaking draws are independent. For random candidate pairs, Spearman's ρ_S compares this produced ranking with expected-output ranks. For selected offers, it compares the produced ranking with realized-output ranks. Equal target values retain their average ranks. Thus a constant predictor produces a random ranking with expected rank correlation zero. In the definitions above, q denotes the relevant target: expected output for random candidate pairs and realized output for selected offers. For selected-offer summaries, R^2 , bias, and rank correlation are undefined when $n < 5$ or $\text{Var}_{\text{pop}}(q) < \sigma_\varepsilon^2/6$, while RMSE is defined whenever $n > 0$. Random-candidate summaries use common valid-agent support for all four measures: an agent-level candidate set that fails the sample-size or variance condition is excluded from the R^2 , RMSE, bias, and rank-correlation means.

Other Measures

Measure	Definition
Access vs. assessment	For each accepted broker-directed offer, record whether receiver j was a direct neighbor of sender i in G at the time of the offer. Non-neighbor receivers indicate access; neighbor receivers indicate assessment.
Realized match output by channel	Average realized match output $\bar{q}_c^t$, where $c \in \{\text{self, brokered}\}$.
Requested-position outcomes	For each channel, record the share of requested positions filled and realized output net of channel costs per requested position. Unfilled positions contribute zero output.
Mean channel satisfaction	Cross-agent means $N^{-1} \sum_i s_{i,\text{self}}^t$ and $N^{-1} \sum_i s_{i,\text{broker}}^t$.
Outsourcing rate	Outsourced requested positions divided by total requested positions.
Broker access size	$\ \mathcal{A}_b^t\ = \ \text{Roster}^t \cup D^t\ $, measured after outsourcing decisions form D^t and before entry/exit.
Counterparty concentration	Median and maximum current-period counterparties per agent.
Agent-node degree summaries	Mean, median, minimum, and maximum degree, excluding the broker node.

Part II. Baseline Values and Reproducibility

9. Baseline Values and Reproducibility

9a. Baseline Parameter Values

The table lists baseline values only. Entries are arranged in two columns; each column follows the document order, with the run length listed first.

The baseline learning function is NN. Ridge and its broker variants are alternate experiments (§3).

The baseline broker provides both access and assessment. The assessment-only and access-only modes are alternate experiments (§6c).

Symbol	Baseline	Defined in	Symbol	Baseline	Defined in
T	500	§1	f_r	0.60	§4a
N	1000	§1	λ_c	0.05	§4c
d	8	§1	k_G	6	§5a
d_γ	8	§1	p_{rewire}	0.1	§5a
σ_x	0.5	§1	η_{exit}	0.02	§5c
Q	2.84	§2a	K	5	§6
ρ	0.50	§2a	p_{demand}	0.50	§6
δ	0.5	§2c	$n_{\text{strangers}}$	10	§6a
σ_ε	0.28	§2a	α_R	0.20	§6b
$\eta_{\text{lr},a}, \eta_{\text{lr},b}$	0.003, 0.03	§3a	p_{roster}	0.02	§6b
$E_{\text{init},a}, E_{\text{init},b}$	100, 100	§3a	ω	0.2	§7a
$E_{\text{steps},a}, E_{\text{steps},b}$	50, 50	§3a	Δ_{net}	20	§8
W_p	40 periods	§3a	M_{train}	2000	§3a

9b. Reproducibility

All model-event randomness flows from a single integer seed. The seed determines type draws, G , matching function parameters ($\mathbf{c}$, $\mathbf{A}$, $\mathbf{B}$), broker seed roster, standing-roster churn and replenishment, random ordering within market score ties, and subsequent model events. Random candidate samples and their independent agent and broker tie-breaking draws are deterministic conditional on seed and period. Simulations are fully reproducible given parameters and seed.

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Supplementary Material

Additional Model Diagnostics and Robustness Analyses

Data-generating process

Figure S1 illustrates the baseline type distribution. Figures S2–S3 describe the match-value function under that distribution. Within each realization, principal types and the matching-function objects c , A , and B are held fixed while ρ and δ vary. The match-value figures show the conditional value $E[q_{ij} | x_i, x_j] = Q + f(x_i, x_j)$, without match noise. They therefore isolate the structure principals and the broker must learn rather than variation from one set of noisy outcomes.

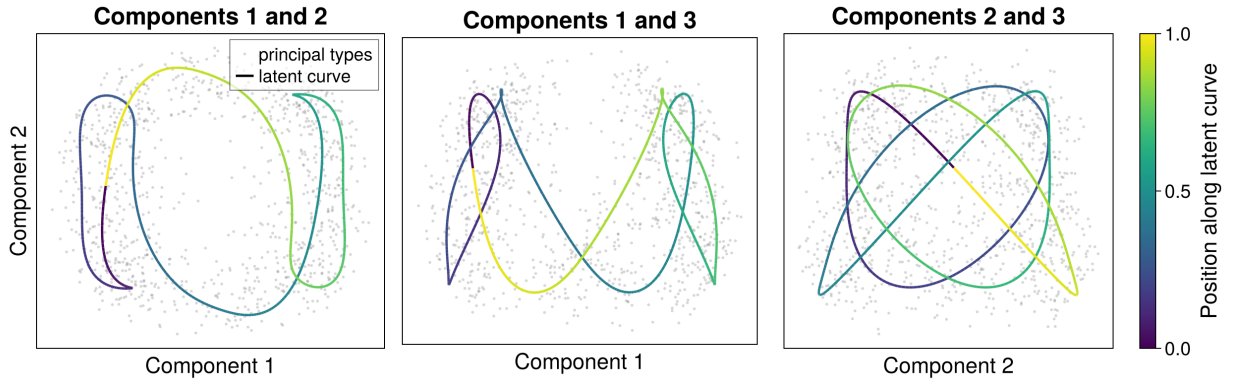


Figure S1: Three views of the baseline type distribution for all 1000 principals in realization 1. Principal types are drawn around a smooth latent curve and projected to the unit sphere in the model’s eight-dimensional type space. For visualization, the figure projects both the realized types and the noiseless curve onto the curve’s first three principal components. Gray points are realized principal types, the colored line is the latent curve, and color records position along that curve. The projection is used only for visualization; matching and learning use the complete eight-dimensional type vectors.

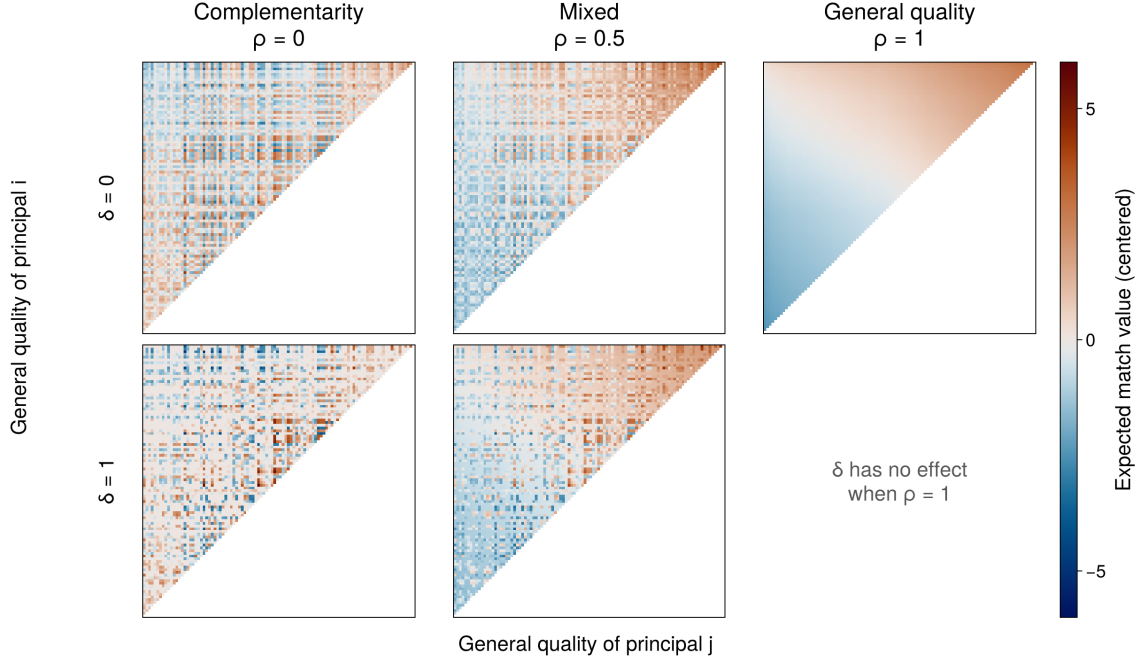


Figure S2: Centered conditional match value for 100 principal types selected at evenly spaced general-quality ranks from a 1000-principal realization (seed 1). Columns vary the share of predictable match-value variation attributed to general quality (ρ); rows compare low difficulty ($\delta = 0$) with high difficulty ($\delta = 1$). Because δ is inactive at $\rho = 1$, that boundary condition is shown once. The realized types and matching-function objects c , A , and B are identical across panels. General quality increases from left to right and from bottom to top. Each symmetric matrix is shown once in its upper triangle; the diagonal is excluded because principals cannot match with themselves. All panels use the same off-diagonal color scale. The matrices exclude random match variation and are centered by subtracting their grand mean. The value scale is normalized using separate reference markets: across those markets, the average within-market variance in predictable match value among possible pairs equals one.

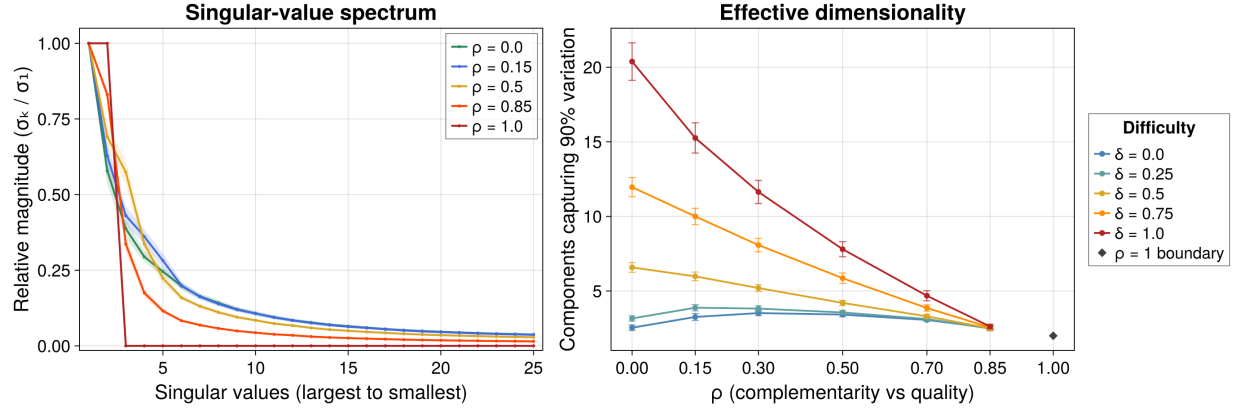


Figure S3: Dimensionality of the centered conditional match-value matrix over 50 independently generated sets of principal types and matching-function objects. Left: the first 25 singular values at $\delta = 0.5$. Each value is divided within its realization by the largest singular value, σ_1 . Lines are means across the 50 realizations and ribbons are pointwise 95% Monte Carlo intervals. Right: within each realization, the minimum number of singular components needed to capture 90% of match-value variation. Markers are means across the 50 realizations and whiskers are 95% Monte Carlo intervals. The pure-quality boundary $\rho = 1$, where δ is inactive, is shown once and unconnected.

Alternative structural measures

Figures S4–S6 reproduce the main structural analyses using the broker’s two other saved ego-network measures, Burt’s aggregate network constraint and Burt’s effective size. Both are recomputed for the broker every 20 periods. As in the main text, a regime is one scientifically distinct parameter configuration; duplicate grid coordinates and inactive parameters do not create additional regimes.

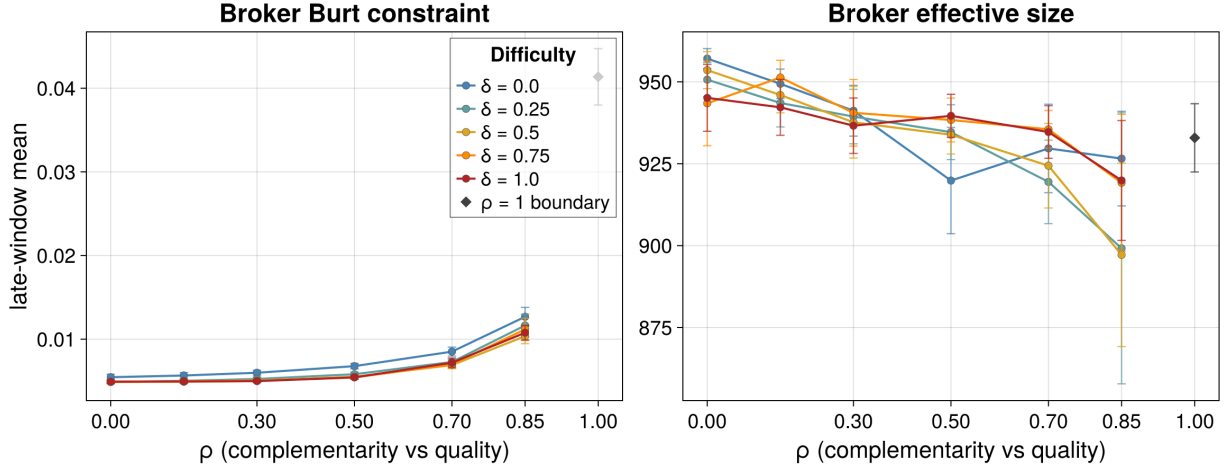


Figure S4: Broker Burt constraint (left) and broker effective size (right) across the $\rho \times \delta$ grid. Lines for each difficulty level δ span $\rho = 0$ through 0.85. The pure-quality boundary $\rho = 1$, where δ is inactive, is shown once and unconnected. Markers are late-window means over seeds and whiskers are 95% Monte Carlo intervals.

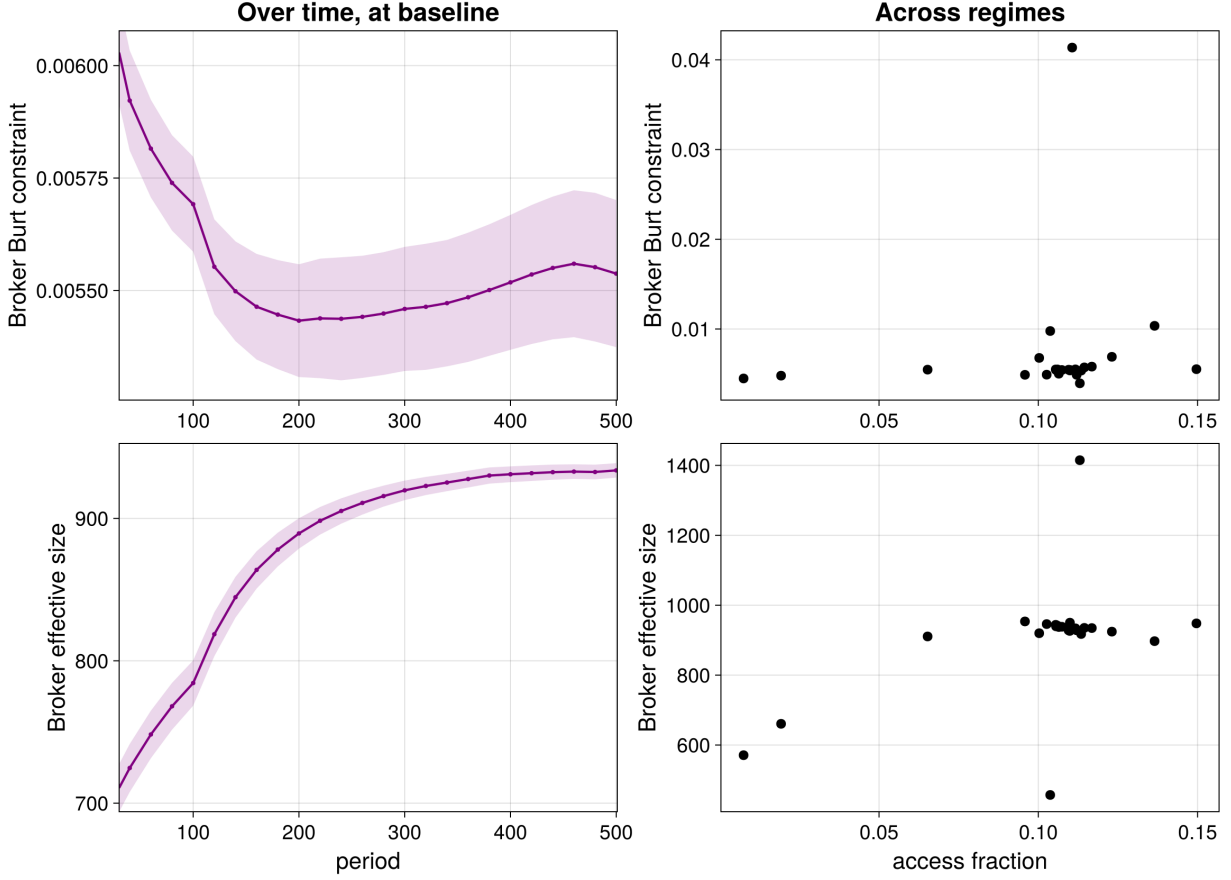


Figure S5: Broker Burt constraint (top) and broker effective size (bottom). Left: each measure over time at the baseline, shown as a 5-observation rolling ensemble mean over seeds with pointwise 95% Monte Carlo intervals. Network measures are computed every 20 periods. Right: each measure against the access fraction, with one late-window regime mean per one-at-a-time regime. Time starts at $t = 30$ to omit the initialization transient.

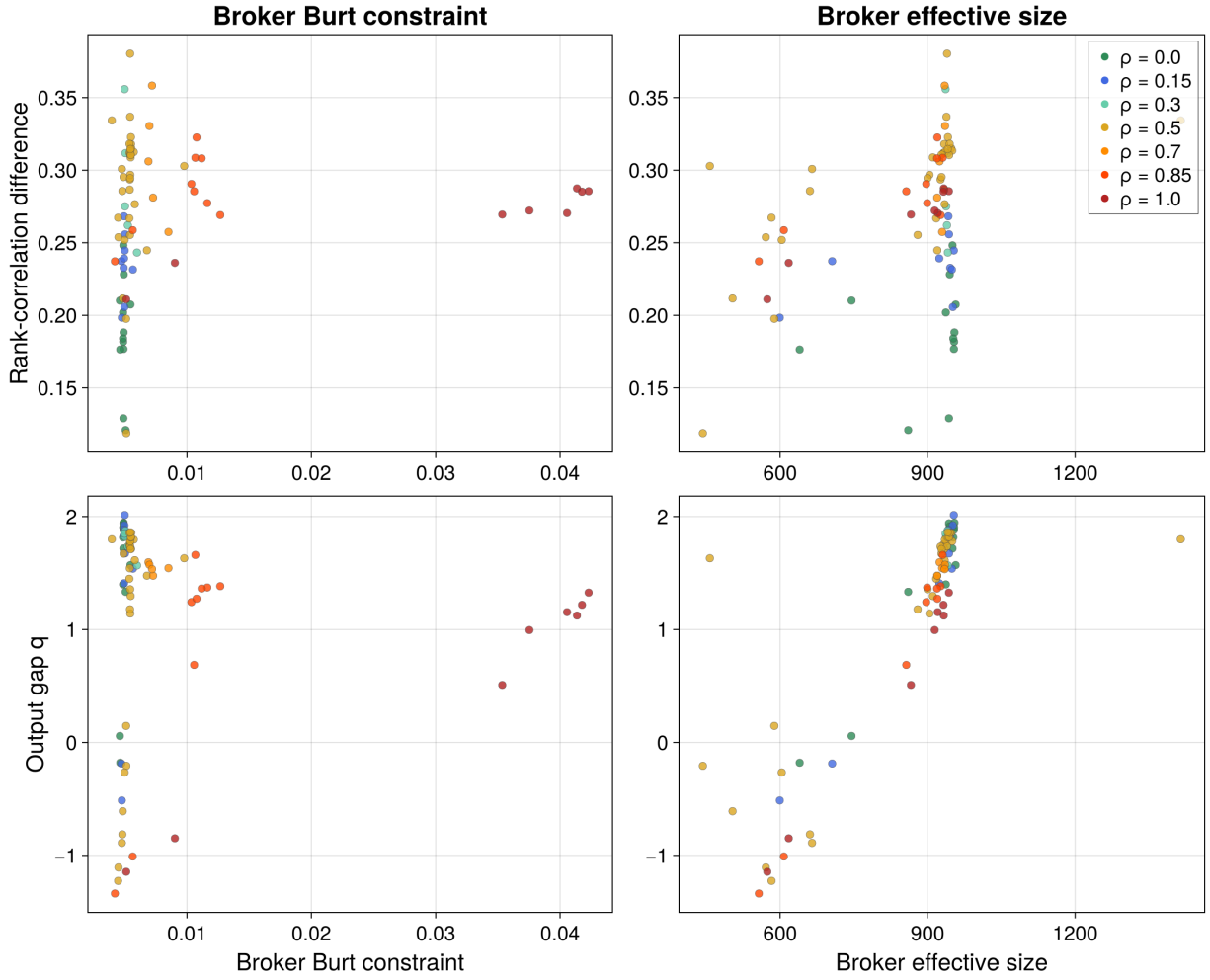


Figure S6: Difference in rank correlation (broker minus principal, top) and output gap (broker channel minus self-search, bottom) against broker Burt constraint (left) and broker effective size (right). Each point is one regime's late-window mean across seeds, colored by ρ .